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ORIGINAL ARTICLE

Genome-wide identification and characterization of the *bZIP* gene family and their expression in emmer wheat (*Triticum turgidum* subsp. *dicoccum*)

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Abstract
As one of the most abundant and conserved gene families in eukaryotes, *bZIP* transcription factors are essential for plant development, growth, and responses to abiotic stress, as well as anthocyanin accumulation. This study investigates the basic leucine zipper (*bZIP*) transcription factor gene family in Emmer wheat (*Triticum turgidum subsp*. *dicoccum*), an ancient grain recognized for its nutritional benefits and agricultural resilience. In this study, we identified 70 *bZIP* genes that were unevenly distributed in the emmer wheat genome and named them *TdbZIP1* to *TdbZIP70*. These *TdbZIP*s were clustered into six clades with clade-specific motifs and structures. We examined exon–intron structures, conserved motifs, and cis-regulatory elements, providing insights into the functional roles of these genes. Synteny analysis illustrated the conservation of gene order among related species, while Circos plots visualized complex genomic relationships. Additionally, tissue-specific expression analysis highlighted the differential expression of *bZIP* genes across various tissues. The whole-genome duplication (WGD) analysis suggested that segmental duplication is likely the primary factor contributing to the expansion of the *TdbZIP* family. This research represents the first report on the *bZIP* gene family in emmer wheat, contributing to a deeper understanding of their roles in crop improvement, sustainable agriculture, and food security.

Keywords Basic leucine zipper (*bZIP*) · Transcription factors · Gene family · Emmer wheat · *TdbZIP* genes · Tissue-specific expression

Abbreviations

ORF	Open reading frame
HMM	Hidden Markov model
SMART	Simple modular architecture research tool
BLAST	Basic local alignment search tool
MEGA	Molecular evolutionary genetics analysis
CDD	Conserved domain database
GSDS	Gene structure display server
MEME	Multiple expectation maximization for motif elicitation
pI	Isoelectric point
ABRE	Abscisic acid responsive element
ARE	Auxin responsive element
LTRE	Low-temperature responsive element
MeJA	Methyl jasmonic acid-responsive element
GARE	Gibberellin-responsive element
DTRE	Drought-responsive element

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Introduction

Emmer wheat (*Triticum turgidum subsp. dicoccum*) is one of the world's oldest domesticated crops, and it harbors a potentially rich reservoir of agronomic and nutritional quality trait variations (Fadida-Myers et al. 2022). It is known for its rich nutritional profile, including high levels of protein, fiber, and essential minerals, which contribute to better blood sugar control and improved digestion. It also contains antioxidants that support heart health and reduce inflammation. As a hardy and pest-resistant crop, emmer thrives in poor soils with minimal chemical use, making it suitable for sustainable farming practices. Additionally, it serves as a valuable genetic resource for breeding stronger wheat varieties, enhancing food security and climate resilience in modern agriculture. The growing global demand for plant-based health foods has sparked renewed interest in emmer wheat, highlighting its potential as a source of agronomic and nutritional quality trait variations.

Transcription factor (TF) play an indispensable role in plant growth, development, and resistance to biotic and abiotic stresses (Wang et al. 2022). Several families of transcription factors have been identified both in animals as well as plants. The basic leucine zipper (*bZIP*) transcription factor is one of the most abundant and conserved gene families in eukaryotes. The members of the *bZIP* TF family harbor a highly conserved, 60–80 amino acid long domain, which is composed of a basic region and a leucine zipper region. The sequence of the basic region consists of approximately 20 amino acid residues with a fixed nuclear localization structure N-×7-R/K, which can specifically bind to DNA cis-elements (Lee et al. 2006; Nijhawan et al. 2008). The leucine zipper region, containing the core sequences of L-×6-L-×6-L, consists of various repetitions of leucine or other hydrophobic amino acids, which facilitates hetero- or homo-dimerization of *bZIP* proteins (Jakoby et al. 2002). In addition to participating in plant development and growth, *bZIP* transcription factors play crucial roles in various abiotic stress responses and anthocyanin accumulation.

In recent studies, a large number of *bZIP* genes have been found to play important roles in the processes of plant growth and development, such as seed maturation and germination (Toh et al. 2012), flower development (Strathmann et al. 2001), vascular development (Gibalová et al. 2017), and embryogenesis (Guan et al. 2009). For example, the *AtbZIP11* affects plant root development by linking low-energy signals to auxin-mediated control of primary root growth (Weiste et al. 2017). Overexpression of the *ZmbZIP4* in maize can also lead to an increase in the number of lateral roots, longer primary roots, and improved plant roots (Ma et al. 2018). In addition, the *bZIP* genes also play an important role in plant biotic and abiotic stress (Yoshida et al. 2010; Wu et al. 2018; Gai et al. 2020). In wheat, *TabZIP15* promotes the combination of ABF/AREB and ABRE (ABA response element) cis-acting elements through the expression of ABF/AREB to induce downstream target gene expression responding to plant salt and drought stress (Bi et al. 2021). Similar results were observed for the *GsbZIP67* gene in Alfalfa (*Medicago sativa*), overexpression of *GsbZIP67* promoted the growth of plant roots and shoots and changed the physiological indicators of transgenic plants under bicarbonate salt-alkali stress (Wu et al. 2018).

Many *bZIP* TF families have been identified in important plant species. For example, 75 *bZIP* genes have been predicted in *Arabidopsis thaliana* (Jakoby et al. 2002). One hundred twenty-five members of the *bZIP* genes family were identified in maize (*Zea mays*) (Wei et al. 2012a, b), 89 in rice (*Oryza sativa*) (Nijhawan et al. 2008), 114 in apple (*Malus domestica*) (Li et al. 2016), 92 in pear (*Pyrus bretschneideri*) (Ma et al. 2021), 77 in tobacco (*Nicotiana tabacum*) (Duan et al. 2022), 65 in pomegranate (*Punica granatum*) (Wang et al. 2022), and 227 in wheat (*Triticum aestivum*) (Liang et al. 2022a, b). However, there are no reports on the identification and functionality of *bZIP* genes in emmer wheat (*Triticum turgidum* subsp. *dicoccum*).

In this study, we aimed to identify *bZIP* genes in emmer wheat using an in-house assembled whole genome. We conducted phylogenetic analysis, conserved motif characterization, and gene structure assessment to investigate relationships among the identified *TdbZIP* family members.

Materials and methods

This section outlines the comprehensive approach used to analyze the *bZIP* gene family in *Triticum turgidum subsp. dicoccum* (emmer wheat). It begins with the assembly of the emmer wheat genome at the chromosome level, followed by the identification and validation of *bZIP* genes using HMMER and SMART tools. Subsequent analyses include multiple sequence alignment, phylogenetic tree construction, gene structure characterization, and conserved motif prediction. The chromosomal distribution of *bZIP* genes was mapped, and synteny and collinearity with *Arabidopsis thaliana* were examined to explore evolutionary relationships. Promoter analysis identified regulatory elements associated with hormone responses and abiotic stress. The physicochemical properties and subcellular localization of *TdbZIP* proteins were also determined. Finally, RNA-seq data from multiple tissues was analyzed to profile the expression patterns of Td*bZIP* genes, providing insights into their functional roles in emmer wheat.

Genome sequence data retrieval

The genome sequence data for emmer wheat (*Triticum turgidum subsp. dicoccum*) commonly known as emmer wheat, utilized in this study, was derived from an in-house assembly, representing the first assembly of this subspecies. The genome was assembled at the chromosome level to create a comprehensive resource for further analyses. The genome sequence of emmer wheat is currently deposited at NCBI under the accession number SRR33557405, associated with BioProject PRJNA1262668 and BioSample SAMN48478886.

Identification of *bZIP* family members

To identify *bZIP* gene family members, we used the hidden Markov model (HMM) profile specific to the *bZIP* domain (PF00170) from the Pfam database (Punta et al. 2012). The HMMER3.0 (Eddy 2011) was used to identify sequences containing the *bZIP* domain (PF00170) in the emmer wheat genome. Initially, all hits were screened for the *bZIP* domain, and redundant sequences were removed. For further validation, the simple modular architecture research tool (SMART) (Letunic et al. 2021) confirmed the presence of the *bZIP* domain in each candidate. Final *bZIP* gene candidates were assigned names based on their chromosomal locations (e.g., *TdbZIP1*, *TdbZIP*2, etc.). The HMM profile specific to the bZIP domain was used with an E value threshold < 1e-5.

Multiple sequence alignment and phylogenetic analysis

The full-length amino acid sequences of the identified *TdbZIP* proteins were aligned using the MUSCLE algorithm (Edgar 2004). A phylogenetic tree was constructed with MEGA-11.0 software (Tamura et al. 2021) using the neighbor-joining method and 1,000 bootstrap replicates to assess tree reliability. The Interactive Tree of Life (iTOL) v5.0 (Letunic and Bork 2021) was used to visualize the phylogenetic relationships between *TdbZIP* genes and their *Arabidopsis thaliana* orthologs.

Gene structure and motif analysis

The exon–intron structures of *TdbZIP* genes were analyzed with the gene structure display server (GSDS 2.0) (Hu et al. 2015) by comparing coding sequences to their corresponding genomic sequences to determine exon and intron counts. Conserved motifs within TdbZIP proteins were predicted using the multiple expectation maximization for motif elicitation (MEME) suite (Bailey et al. 2015), with the maximum number of motifs set to 10. The visualization of the conserved motifs was carried out using TBtools (Chen et al. 2020).

Chromosomal location of *TdbZIP* genes

The chromosomal distribution of *TdbZIP* genes was visualized using MG2C (multiple genome comparison and visualization tool) (Chao et al. 2021). This web-based tool facilitates the generation of chromosome maps that highlight the spatial organization of *TdbZIP* genes across the fourteen chromosomes of the emmer wheat genome. Gene positions were plotted to examine their physical locations, focusing on regions with higher gene densities that may indicate gene duplication events or evolutionary hotspots. This plotting was done proportionally based on actual chromosomal lengths using coordinates derived from genome annotation.

Promoter sequence and Cis-acting element analysis

To explore the regulatory functions of *TdbZIP* genes, the 2000 bp upstream promoter regions of each gene were extracted from the emmer wheat genome. These promoter sequences were subjected to analysis for cis-acting regulatory elements using the PlantCARE database (Lescot 2002). The analysis focused on identifying cis-elements linked to hormone responses, specifically abscisic acid (ABA), salicylic acid (SA), gibberellins (GA), and methyl jasmonate (MeJA), as well as elements associated with abiotic stresses such as drought, light exposure, and low temperatures. Visualization of these elements was performed using TBtools software (Chen et al. 2020). Motifs with incorrect assignments (e.g., overlaps, truncations) were filtered using custom scripts prior to visualization.

Physicochemical properties and subcellular localization of *TdbZIP* proteins

The physicochemical characteristics of *TdbZIP* proteins, such as molecular weight and isoelectric point (pI), were determined using the NaturePred tool (Sharanbasappa et al. 2024). Additionally, the subcellular localization of these proteins was predicted through the WoLF PSORT algorithm (Horton et al. 2007), which analyzes amino acid sequences to infer potential protein localization within the cell.

Synteny and collinearity analysis of Td*bZIP* genes

The analysis of synteny and collinearity between the *TdbZIP* genes of emmer wheat (*Triticum turgidum* subsp. *dicoccum*) and those from *Arabidopsis thaliana*, *Triticum aestivum, Hordium vulgare, Triticum turgidum subsp. durum*, and *Triticum turgidum subsp. dicoccoides*. Initially, *TdbZIP* protein sequences from emmer wheat and five other species were aligned using the diamond blastp tool (Buchfink et al. 2015) to identify potential orthologous genes. This bidirectional blastp search involved using the *TdbZIP* proteins from one species as queries against the *bZIP proteins of other species.* This approach aimed to uncover reversible best hits (RBH), which are indicative of likely orthologous relationships between the two species. Following the identification of RBH pairs, genomic coordinates were extracted from the GFF3 files of both species utilizing the rtracklayer (Lawrence et al. 2009), and GenomicRanges packages (Aboyoun et al. 2013).

Circos plot and duplication analysis

To analyze tandem and segmental duplications in the emmer wheat genome, we created a Circos plot using the circlize package in R (Gu et al. 2014). Tbtools was used to calculate estimate Ka (non-synonymous), Ks (synonymous), and their ratios for pPairs from segmental or tandem duplications.

RNA sample preparation, illumina sequencing, and RNA-seq data processing

A total of three independent samples of emmer wheat were collected for each of the four tissue types, viz. stem, flag leaf, seedling-15 days, and root. A total of 12 RNA samples (three biological replicates for four tissue type) was isolated using PureLink™ RNA Mini Kit (Thermofisher Scientific) following the provided protocol. RNA was quantified and quality assessed with Nanodrop. Then, for each tissue type equal concentration was taken from the three biological replicates and pooled into one. The libraries were sequenced on the Illumina Novaseq 6000 platform. Raw RNA-seq data for four tissue types was subjected to quality assessment using FastQC (Andrews S. 2010) to ensure data integrity. Low-quality reads (LEADING: 3, TRAILING:3) and adapters were removed using Trimmomatic (Bolger et al. 2014) using default parameters (ILLUMINACLIP: TruSeq3-PE.fa:2:30:10, SLIDINGWINDOW:4:15, MINLEN:36). The RNA-seq data used in our study is currently deposited at the NCBI with BioProject PRJNA1260914.

Expression profiling of *TdbZIP* genes in emmer wheat (*Triticum turgidum* subsp. *dicoccum*) across different tissues

To investigate the *bZIP* gene family in emmer wheat across different tissues, a comprehensive RNA-seq analysis was conducted. The overall workflow for identifying and characterizing the bZIP gene family is illustrated in Fig. 1. Cleaned reads of each tissue type were aligned to the Emmer wheat (*Triticum turgidum* subsp. *dicoccum*) reference genome using BWA for accurate mapping (Li And Durbin 2009). The alignment files in SAM format were converted to BAM and subsequently sorted with Samtools (Li et al. 2009) for downstream analysis. Gene expression quantification was performed on the sorted BAM files using feature count. Manual fold change approach was adopted to identify differentially expressed genes (DEGs) across tissue pairs. Raw read counts were normalized using a log2 transformation with a pseudocount of 1. Genes were considered differentially expressed based on empirical thresholds: |log2 fold change|> 1 with pseudo p values of 0.01 or lower, and |log2

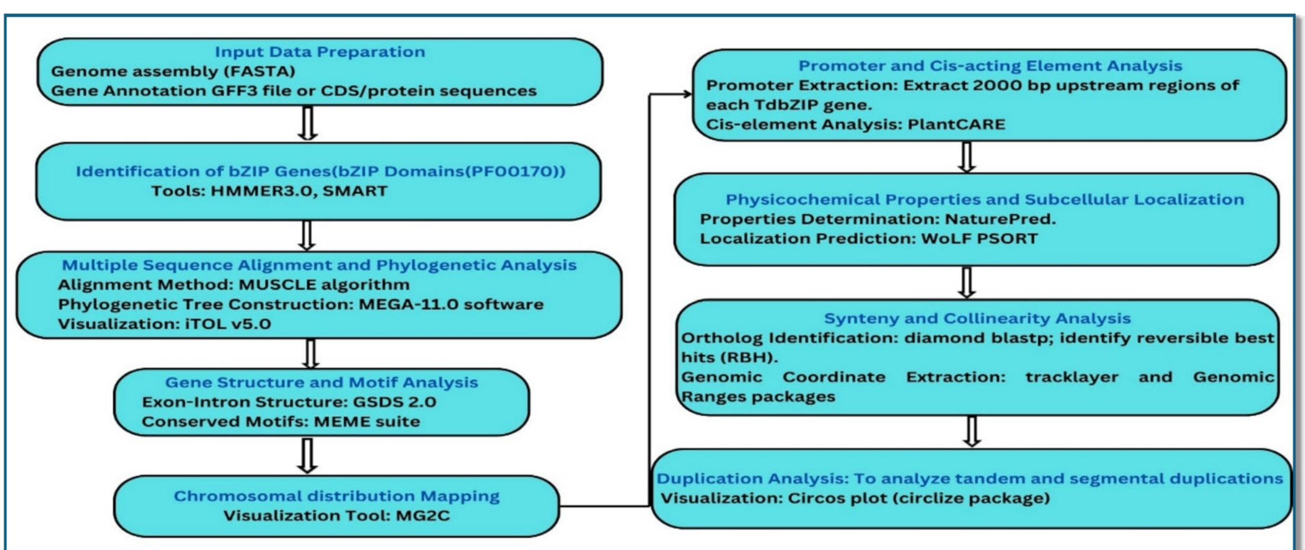

Fig. 1 Flowchart: Steps for identifying and characterizing the bZIP gene family in *Triticum turgidum subsp. dicoccum*

fold change|> 2 with pseudo p values of 0.001. While this method does not model biological variability, it provides a comparative snapshot of gene regulation across tissues under limited replicates. The workflow for tissue-specific expression analysis is summarized in Fig. 2. DEG lists were generated for all six pairwise comparisons among Stem, Root, Flag Leaf, and 15-day Seedling tissues.

Results

Genome-wide identification of *bZIP* family members in *T. dicoccum*

We identified 70 *bZIP* genes in the *T. dicoccum* genome, named *TdbZIP1* to *TdbZIP70* according to their order of localization on the chromosomes (Table 1). The lengths of *TdbZIP* mRNA protein sequences ranged from 92 amino acids (*TdbZIP1*) to 869 amino acids (*TdbbZIP32*). The average molecular weight of *TdbZIP* family members was 26,325.36 kDa. The average isoelectric point (pI) of *TdbZIP* genes was 4.28 (*TdbZIP70*) to 10.250 (*TdbZIP61*). The proteins with an isoelectric point less than 7 accounted for 80% of the total number, which means most of the TdbZIP proteins were acidic. A plot of the molecular weight with pI for each *TdbZIP* gene revealed that the some of *TdbZIPs* clustered together, indicating that they have a similar properties and some are scattered (Fig. 3). The grand average of hydropathy index (GRAVY) values for *TdbZIP* members ranged from -1.168 to -0.325, suggesting that these proteins are hydrophilic.

Phylogenetic analysis, classification, and multiple sequence alignment of *TdbZIP* proteins

To investigate evolutionary relationships, a neighbor-joining phylogenetic tree was constructed using full-length protein sequences of *bZIP* genes from *T. dicoccum* and *Arabidopsis* (Fig. 4). The phylogenetic analysis reveals the evolutionary relationships among 70 *TdbZIP* proteins and 57 *Arabidopsis thaliana bZIP* (*AtbZIP*) proteins, categorized into six subfamilies (A–F). Subfamily B is the largest, comprising 37 *TdbZIP* members alongside 10 At*bZIP* proteins, suggesting significant gene expansion in *T. dicoccum*. They might play a role in vascular development. Subfamily A contains 25 *TdbZIP* and 1 *AtbZIP* proteins, reflecting strong evolutionary conservation transcriptionally activated after stress treatment (e.g., cold, drought, anaerobiosis, wounding). (Jakoby et al 2002) Subfamily C is the smallest sized, with 0 *TdbZIP* and 4 *AtbZIP* proteins, indicating potential functional divergence. Smaller subfamilies D, E, and F include 3, 0, and 4 *TdbZIP* proteins, and 12, 8, 22 *AtbZIP* proteins, respectively, interspersed with *AtbZIP* proteins, pointing to a balance between

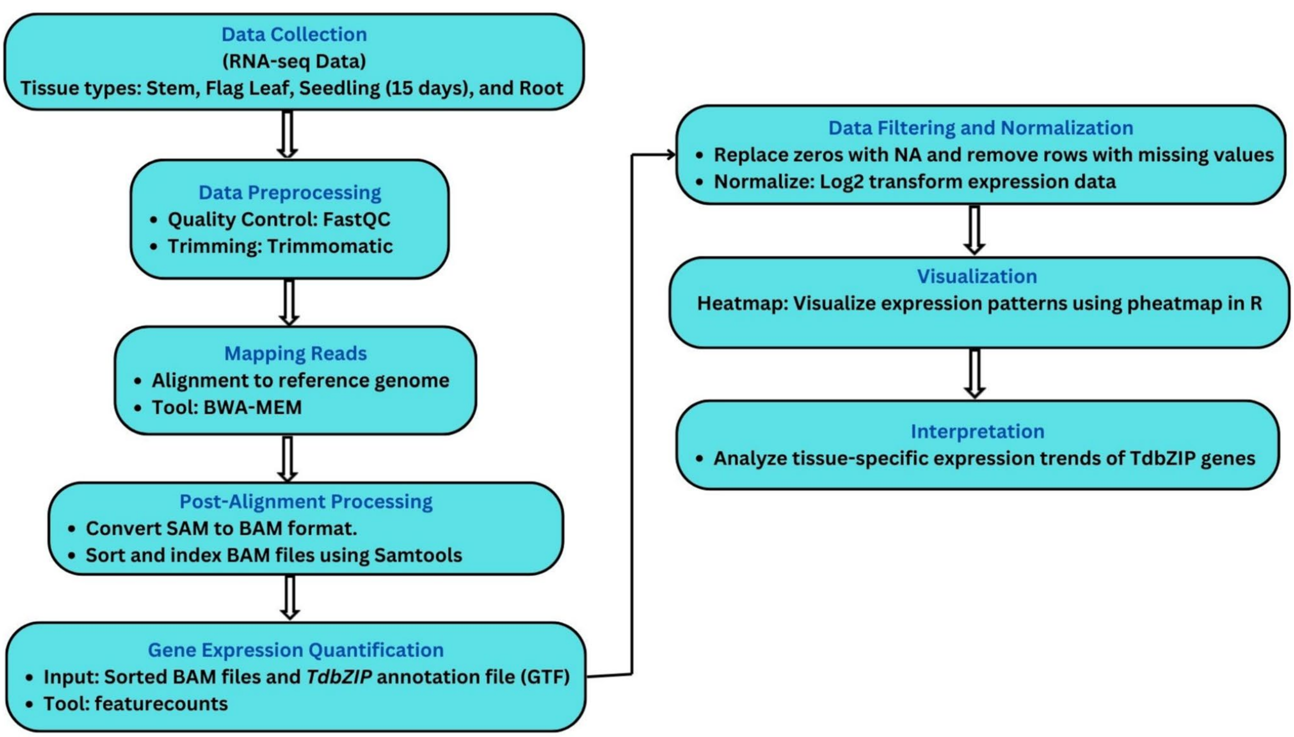

Fig. 2 Flowchart: Tissue-specific expression of TdbZIP genes in emmer wheat (*Triticum turgidum ssp. dicoccum*)

Table 1 Nomenclature and characteristics of the putative basic leucine zipper (*bZIP*) proteins in Emmer wheat (*Triticum turgidum* subsp. *dicoccum*)

Proposed gene name	Gene ID	Chr	start	End	Group	Protein length(aa)	Molecular weight (kDa)	Isoelectric point (pI)	Hydrophobicity (GRAVY)	Predicted subcellular localization
*TdbZIP*1	CM_00001.g18149.t1	1	152,958,495	152,959,245	B	92	10,402.9155	8.829606056	−0.82826087	Nucleus
*TdbZIP*2	CM_00001.g20965.t1	1	176,338,524	176,343,759	B	399	43,868.0901	9.222540855	−0.644862155	Cytoplasm
*TdbZIP*3	CM_00002.g17164.t1	2	141,000,047	141,001,002	B	168	18,733.4088	5.563537025	−0.325	Chloroplast
*TdbZIP*4	CM_00002.g42574.t1	2	341,749,220	341,750,342	B	295	33,965.7301	9.014630318	−1.126101695	Nucleus
*TdbZIP*5	CM_00002.g54584.t1	2	436,973,608	436,974,863	B	263	29,091.681	5.374149513	−0.446007605	Chloroplast
*TdbZIP*6	CM_00002.g55552.t1	2	444,672,534	444,675,621	A	242	27,643.9078	4.865213203	−0.622727273	Cytoplasm
*TdbZIP*7	CM_00002.g65881.t1	2	525,074,355	525,078,162	B	839	92,424.13	5.29559803	−0.660548272	Nucleus
*TdbZIP*8	CM_00003.g4432.t1	3	39,066,357	39,067,450	A	228	25,153.7866	9.159813118	−0.378947368	Chloroplast
*TdbZIP*9	CM_00003.g83892.t1	3	684,683,445	684,686,468	A	242	27,749.0086	5.139233971	−0.695867769	Cytoplasm
*TdbZIP*10	CM_00003.g87591.t1	3	715,008,407	715,011,511	A	242	27,743.9243	4.933306313	−0.680578512	Cytoplasm
*TdbZIP*11	CM_00003.g96087.t1	3	784,844,612	784,845,293	B	132	15,068.8202	6.742377663	−1.168181818	Nucleus
*TdbZIP*12	CM_00004.g6761.t1	4	56,103,947	56,107,099	A	217	24,640.444	4.765631294	−0.592165899	Cytoplasm
*TdbZIP*13	CM_00004.g37463.t1	4	296,307,744	296,308,725	A	113	13,351.0451	5.407059288	−0.607964602	Nucleus
*TdbZIP*14	CM_00004.g37630.t1	4	297,531,491	297,532,223	F	149	16,643.8406	6.3196661	−0.616778523	Nucleus
*TdbZIP*15	CM_00004.g54147.t1	4	432,118,956	432,120,505	B	137	15,106.3468	5.429340172	−0.396350365	Chloroplast
*TdbZIP*16	CM_00004.g55894.t1	4	446,469,383	446,472,475	A	161	18,775.8592	4.586986351	−0.745341615	Cytoplasm
*TdbZIP*17	CM_00004.g56812.t1	4	453,520,145	453,521,532	B	214	23,359.2751	7.841370201	−0.561214953	Chloroplast
*TdbZIP*18	CM_00004.g64415.t1	4	512,803,428	512,805,279	B	460	50,395.0907	4.846740532	−0.468043478	Cytoplasm
*TdbZIP*19	CM_00004.g67416.t1	4	536,288,127	536,288,867	B	151	16,844.026	5.609008217	−0.613245033	Nucleus
*TdbZIP*20	CM_00004.g69497.t1	4	552,713,816	552,715,091	B	332	37,202.8401	5.543074989	−0.465060241	Cytoplasm
*TdbZIP*21	CM_00004.g83989.t1	4	666,402,063	666,405,153	A	129	14,873.3874	4.614837456	−0.927906977	Cytoplasm
*TdbZIP*22	CM_00005.g24279.t1	5	199,991,892	199,992,622	B	149	16,365.378	5.377673531	−0.471812081	Nucleus
*TdbZIP*23	CM_00005.g25409.t1	5	209,677,558	209,680,631	A	161	18,686.6553	4.502864647	−0.817391304	Nucleus
*TdbZIP*24	CM_00005.g32783.t1	5	270,335,244	270,335,952	B	140	15,637.296	6.086114693	−1.014285714	Nucleus
*TdbZIP*25	CM_00005.g41163.t1	5	339,837,747	339,840,813	A	242	27,753.0172	4.956610298	−0.651239669	Cytoplasm
*TdbZIP*26	CM_00005.g45119.t1	5	371,559,406	371,562,467	A	242	27,736.9516	4.99753437	−0.695867769	Cytoplasm
*TdbZIP*27	CM_00005.g62206.t1	5	510,662,979	510,665,743	D	263	28,694.4429	6.05928669	−0.799619772	Nucleus
*TdbZIP*28	CM_00006.g10718.t1	6	87,485,803	87,487,072	B	211	23,710.9906	5.026635933	−0.449763033	Cytoplasm
*TdbZIP*29	CM_00006.g37118.t1	6	290,783,896	290,785,172	F	149	15,951.9346	7.652085686	−0.228187919	Nucleus
*TdbZIP*30	CM_00006.g38243.t1	6	299,528,151	299,529,003	B	157	17,432.7202	4.836623192	−0.407006369	Mitochondria
*TdbZIP*31	CM_00006.g39817.t1	6	311,544,531	311,546,733	A	436	46,895.3628	4.784615517	−0.624770642	Mitochondria
*TdbZIP*32	CM_00006.g54033.t1	6	425,122,734	425,126,532	B	869	96,074.1402	5.595480537	−0.560644419	Nucleus
*TdbZIP*33	CM_00006.g68736.t1	6	539,468,038	539,469,223	B	162	18,143.6233	5.393247414	−0.408641975	Chloroplast
*TdbZIP*34	CM_00007.g39268.t1	7	320,641,723	320,642,343	A	97	11,004.0765	4.260048485	−0.775257732	Cytoplasm
*TdbZIP*35	CM_00007.g88450.t1	7	721,881,846	721,882,490	B	101	11,911.0231	9.931499672	−0.948514851	Cytoplasm/Nucleus
*TdbZIP*36	CM_00008.g42268.t1	8	335,107,226	335,110,276	A	242	27,687.9256	5.127752495	−0.681818182	Cytoplasm

Table 1 (continued)

Proposed gene name	Gene ID	Chr	start	End	Group	Protein length(aa)	Molecular weight (kDa)	Isoelectric point (pI)	Hydrophobicity (GRAVY)	Predicted subcellular localization
*TdbZIP*37	CM_00008.g43176.t1	8	342,295,996	342,299,051	A	242	27,659.9535	4.982528877	−0.641322314	Cytoplasm
*TdbZIP*38	CM_00009.g33392.t1	9	276,878,252	276,880,943	D	239	26,051.1131	5.997161674	−0.446861925	Nucleus
*TdbZIP*39	CM_00010.g795.t1	10	8,368,021	8,368,915	F	210	21,955.5575	9.772842979	−0.301428571	Nucleus
*TdbZIP*40	CM_00010.g11001.t1	10	93,446,054	93,448,515	B	284	31,576.6517	5.494193459	−0.33415493	Chloroplast
*TdbZIP*41	CM_00010.g20139.t1	10	164,639,039	164,640,361	B	371	42,117.1235	5.675680351	−0.6606469	Nucleus
*TdbZIP*42	CM_00010.g20308.t1	10	165,750,452	165,753,572	A	242	27,610.8168	4.974230385	−0.672727273	Cytoskeleton
*TdbZIP*43	CM_00010.g26988.t1	10	217,201,255	217,202,103	B	151	16,721.8197	5.8101614	−0.529139073	Nucleus
*TdbZIP*44	CM_00010.g30422.t1	10	243,339,050	243,342,127	A	242	27,588.8079	4.825369072	−0.640495868	Cytoplasm
*TdbZIP*45	CM_00010.g31061.t1	10	248,381,398	248,384,055	D	239	26,047.1229	5.997161674	−0.453138075	Nucleus
*TdbZIP*46	CM_00010.g34760.t1	10	278,162,294	278,164,477	B	476	50,632.9687	4.564989662	−0.325630252	Cytoplasm
*TdbZIP*47	CM_00010.g39821.t1	10	319,008,574	319,009,247	B	96	11,244.7314	5.465546608	−0.788541667	Cytoplasm/Nucleus
*TdbZIP*48	CM_00010.g60380.t1	10	481,111,177	481,112,254	B	151	16,629.6537	5.510506248	−0.667549669	Nucleus
*TdbZIP*49	CM_00010.g67986.t1	10	540,473,610	540,474,272	B	151	16,984.0594	5.12576313	−0.650331126	Nucleus
*TdbZIP*50	CM_00010.g80651.t1	10	639,131,504	639,133,277	B	407	44,817.4202	7.106431389	−0.522604423	Chloroplast
*TdbZIP*51	CM_00011.g35995.t1	11	295,468,219	295,469,147	A	229	26,332.2851	4.982301521	−0.755895197	Cytoplasm
*TdbZIP*52	CM_00011.g35999.t1	11	295,482,079	295,482,886	A	148	17,316.2865	4.558794212	−0.805405405	Cytoplasm
*TdbZIP*53	CM_00011.g37976.t1	11	311,000,752	311,001,654	B	139	15,605.0752	5.351016045	−1.077697842	Nucleus
*TdbZIP*54	CM_00011.g43005.t1	11	350,934,933	350,935,635	B	139	15,640.2949	6.304603767	−1.041726619	Nucleus
*TdbZIP*55	CM_00011.g55869.t1	11	453,413,322	453,415,865	B	265	29,766.0209	8.265507698	−0.43509434	Cytoplasm
*TdbZIP*56	CM_00012.g12681.t1	12	103,979,377	103,981,136	B	228	25,248.5875	5.254276085	−0.436403509	Chloroplast
*TdbZIP*57	CM_00012.g28629.t1	12	230,327,705	230,328,590	F	177	19,252.9518	7.860323906	−0.885310734	Nucleus
*TdbZIP*58	CM_00012.g55928.t1	12	446,248,338	446,249,570	B	226	25,584.0889	5.088022041	−0.809292035	Nucleus
*TdbZIP*59	CM_00012.g66239.t1	12	527,089,389	527,090,487	F	207	22,988.4693	6.503824425	−0.533333333	Nucleus
*TdbZIP*60	CM_00013.g34169.t1	13	290,541,070	290,544,156	A	161	18,597.7359	4.61910038	−0.739751553	Cytoplasm
*TdbZIP*61	CM_00013.g41461.t1	13	349,941,102	349,943,490	B	260	28,902.0835	10.25016689	−1.031538462	Nucleus
*TdbZIP*62	CM_00013.g60369.t1	13	504,369,778	504,372,823	A	242	27,752.9741	4.881014442	−0.662809917	Cytoplasm
*TdbZIP*63	CM_00013.g61048.t1	13	510,404,290	510,407,404	A	242	27,735.0019	5.112462807	−0.701652893	Cytoplasm
*TdbZIP*64	CM_00013.g66005.t1	13	550,966,790	550,969,880	A	242	27,896.1593	5.06727581	−0.703305785	Cytoplasm
*TdbZIP*65	CM_00013.g73343.t1	13	611,995,036	611,995,783	A	166	19,496.5143	4.417890358	−0.863253012	Cytoplasm
*TdbZIP*66	CM_00014.g26036.t1	14	207,879,452	207,882,732	B	562	61,615.6329	5.121557045	−0.61227758	Chloroplast
*TdbZIP*67	CM_00014.g37122.t1	14	293,909,923	293,910,669	B	151	17,072.1549	5.682330513	−0.713907285	Nucleus
*TdbZIP*68	CM_00014.g37417.t1	14	296,325,352	296,326,089	B	151	16,781.7282	5.562400246	−0.721192053	Nucleus
*TdbZIP*69	CM_00014.g52546.t1	14	415,823,212	415,824,089	B	121	13,474.0816	6.075372124	−0.472727273	Nucleus
*TdbZIP*70	CM_00014.g76222.t1	14	603,705,615	603,706,440	A	98	11,356.2651	4.28170414	−1.059183673	Cytoplasm

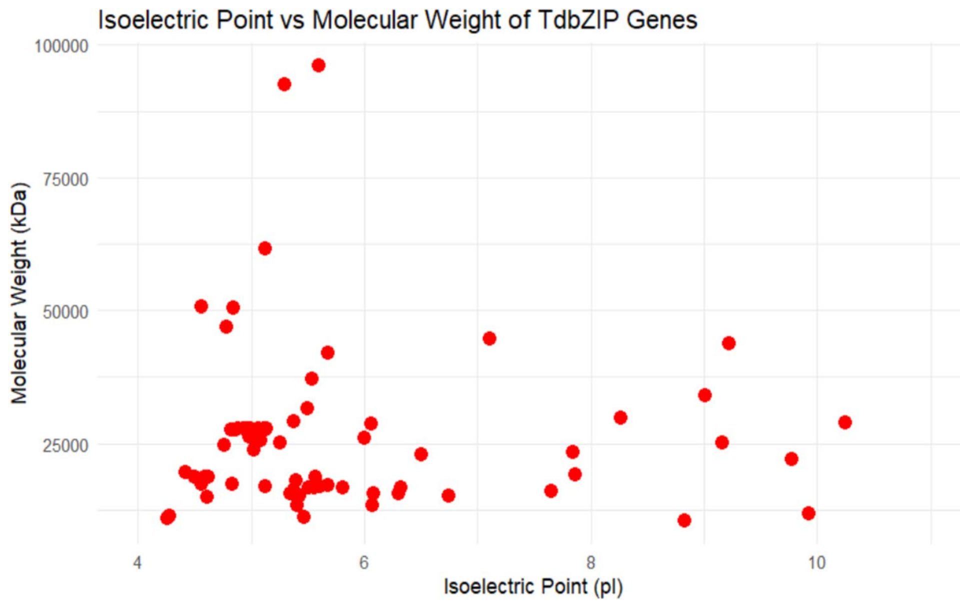

Fig. 3 A plot between isoelectric point vs molecular weight (kDa) for Td*bZIP* genes

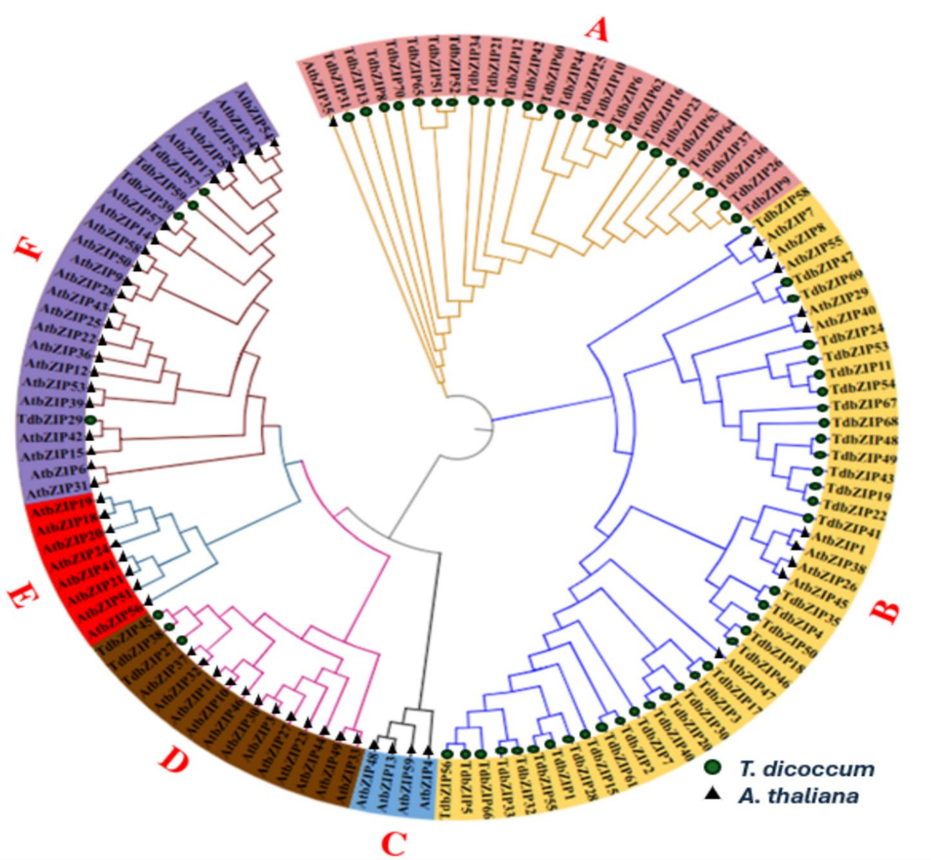

Fig. 4 Phylogenetic analysis of *TdbZIP* proteins of *T. dicoccum* and *Arabidopsis* using neighbor-joining tree by the maximum likelihood method. Different groups are marked with different colors

conservation and specialization. The notable expansion in subfamily B and the varied distribution across other subfamilies suggest that *TdbZIP* genes have undergone adaptive evolution, likely to support specific environmental responses or developmental processes in wheat (Fig. 5).

Gene location on chromosome

Based on genome annotation data, the chromosomal distribution of *TdbZIP* genes was visualized using MG2C_v2.1 (Chao et al. 2021) (http://mg2c.iask.in/mg2c_v2.1/) in

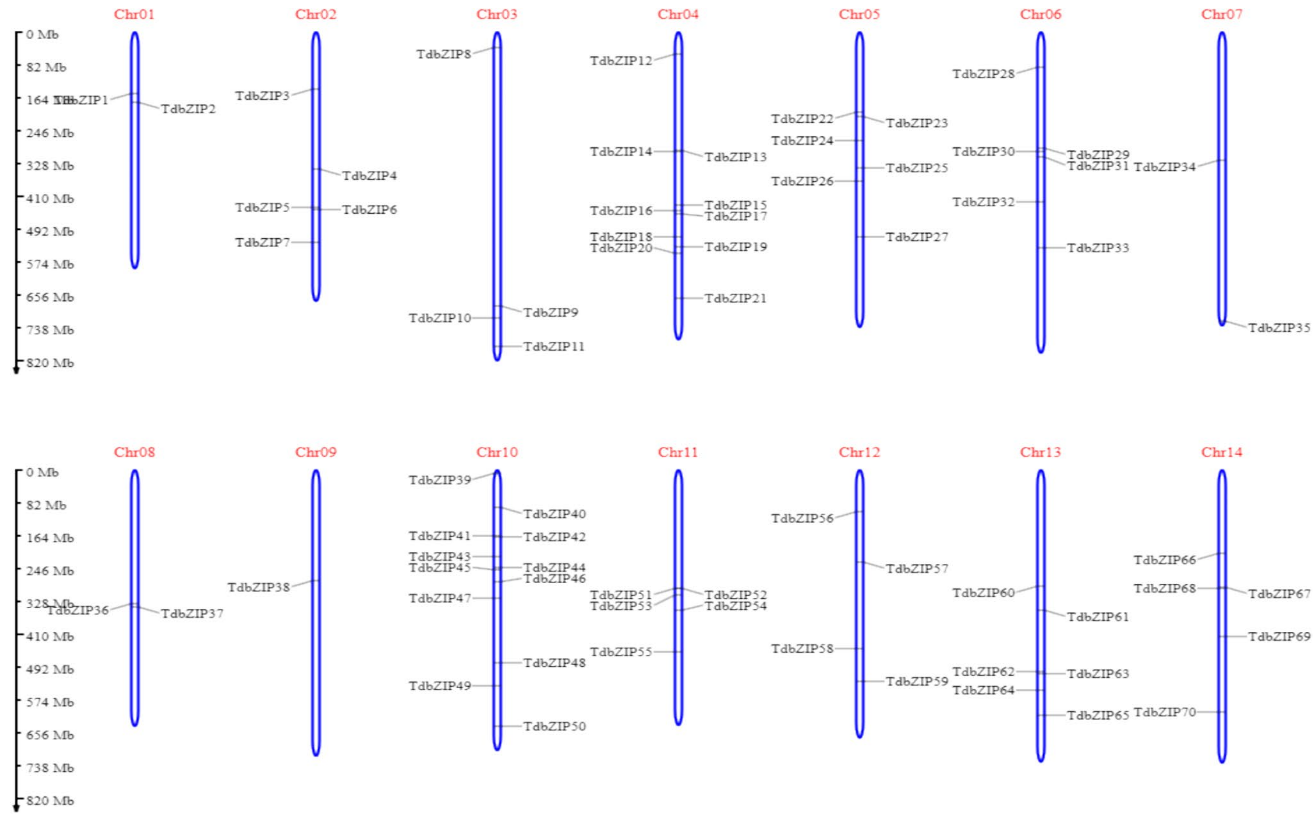

Fig. 5 Chromosomal distribution of *TdbZIP* genes in *T. dicoccum*. The *TdbZIP* genes are marked at their approximate positions on both sides of each chromosome, with chromosome numbers labeled above each bar

T. dicoccum. The *bZIP* genes are distributed unevenly on 14 chromosomes of *T. dicoccum*. A total of 2, 5, 4, 10, 6, 6, 2, 2, 1, 12, 5, 4, 6, and 5 *TdbZIP*s were distributed on Chr1–Chr14, respectively. Chr9 contains the fewest with only 1, while Chr10 contains the most genes with 12 followed by Chr4 with 10 (Fig. 5).

Analyses of gene structure and conserved motifs

In the investigation of the *bZIP* gene family in *Triticum dicoccum*, both the intron–exon organization and motif composition were analyzed. The analysis revealed that the *TdbZIP* genes exhibit a range of exon counts, varying from 1 to 14 exons, with the majority containing between 1 and 3 introns (Fig. 6). Notably, several genes—including *TdbZIP1*, *TdbZIP3*, *TdbZIP11*, *TdbZIP20*, *TdbZIP22*, *TdbZIP24*, *TdbZIP28*, *TdbZIP30*, Td*bZIP35*, *TdbZIP39*, *TdbZIP41*, *TdbZIP*43, *TdbZIP48*, *TdbZIP49*, *TdbZIP51*, *TdbZIP52*, *TdbZIP53*, *TdbZIP54*, and *TdbZIP59*—are completely intron-less. Conversely, the gene *TdbZIP2* was found to have the highest intron count of 13. In addition to gene structure, the conserved motifs within the TdbZIP proteins were identified using the MEME Suite, revealing 10 motifs (Fig. 7). Motif analysis revealed that motif 5 and motif 1 appear to be core conserved sequences in TdbZIP gene family, suggesting that they might have important roles in structural stability and gene function. In addition, motif 5 mostly occurred in subfamily B. Interestingly, group of motif 7, 4, 6, 1, 2, and 8 were only present in subfamily A. Other motifs, such as motifs 2 and 3, were found in specific subgroups, suggesting subgroup-specific functions. Motif 2 corresponds to phosphorylation-related sites, whereas motifs 3 and 5 are associated with DNA-binding and protein–protein interaction domains, as identified by InterProScan. The variation in motif presence and arrangement highlights the functional diversity and evolutionary adaptations of the TdbZIP gene family in *Triticum dicoccum* (Fig. 8).

Promoter region analysis of *TdbZIP* genes

Cis-acting regulatory elements (CAREs) are short DNA sequences in the promoter regions of genes that serve as binding sites for transcription factors, thereby regulating gene expression in response to developmental cues and environmental stimuli. To understand the regulatory mechanisms of *TdbZIP* genes, we analyzed the 2000 bp upstream regions using the PlantCARE webserver, identifying 28 unique CAREs associated with light

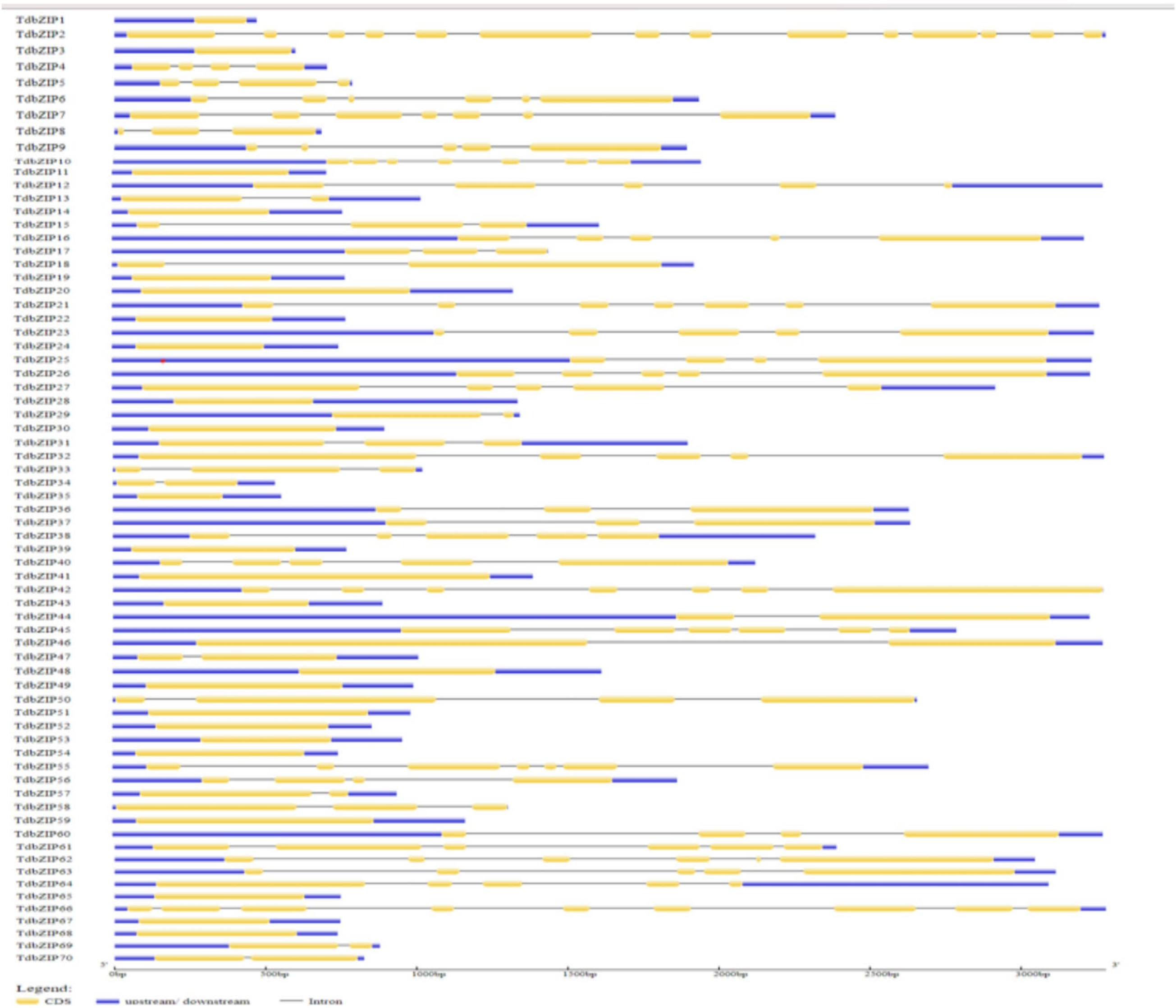

Fig. 6 DNA structures of the *bZIP* gene family in *T. dicoccum*. Exons are indicated by yellow bars and introns are denoted by black lines

responsiveness, hormonal regulation, and stress responses (Fig. 9a, b, c). Key elements, such as G-box, ABRE, MBS, and TGACG-motif, were widely distributed across the genes, indicating their roles in light signaling, abscisic acid response, drought stress, and defense pathways. Elements related to phytohormones (e.g., salicylic acid, gibberellin, and auxin) and stress-responsive motifs like TC-rich repeats and LTR further highlight the involvement of *TdbZIP* genes in integrating environmental and endogenous signals. Additionally, developmental elements, such as chs-Unit 1 m1 and O2-site, suggest their roles in seed-specific regulation and flavonoid biosynthesis. The differential distribution and clustering of these CAREs across the *TdbZIP* genes reveal their complex regulatory potential, enabling plants to adapt to varying environmental conditions while coordinating growth and development.

Subcellular localization of *TdbZIP* proteins

The subcellular localization of all revealed diverse functional roles among the 70 genes. Highest *TdbZIP* genes (29) were localized in the nucleus Rest of the *TdbZIP* were located in cytoplasm (26), chloroplast (10), a smaller number of genes were found to be localized in both the cytoplasm and nucleus (2), as well as in the mitochondria (2) and cytoskeleton (1) (Table 1). Most of the Tfs are usually localized in nucleus and are involved in transcriptional regulatory role (Dong et al 2023). However, some bZIPs exhibit diverse localization and translocate to nucleus in response to environmental stress (ABA, drought stress etc.) or specific cellular signals (Liu et al 2007).

Fig. 7 Sequence logos and legend for the 10 conserved motifs identified in *TdbZIP* proteins

Duplication and collinearity analysis of *TdbZIP*

Duplication and collinearity relationships among *TdbZIP* genes in *Triticum dicoccum* highlighted extensive gene duplications across its genome (Fig. 10). Genes, like *TdbZIP*36 (chromosome 8) and TdbZIP62 (chromosome 13), have multiple connections across different chromosomes, suggesting that they are part of conserved genomic blocks that have undergone duplication and are now spread across the genome. This pattern reveals hotspots of *TdbZIP* gene expansion, likely due to evolutionary events such as segmental or whole-genome duplications, which may have diversified gene functions. Such expansions of the *TdbZIP* family could be crucial for understanding regulatory roles and stress response mechanisms in *T. dicoccum*. Further analysis of evolutionary timeline into 19 paralogous gene pairs identified via phylogenetic analysis revealed a wide range of duplication events, with estimated duplication times spanning from approximately 3.76 MYA to 230.41 MYA (Table 7). Notably, the majority of these duplications appear to be relatively recent (< 50 MYA), suggesting that they most likely originated from small-scale events such as tandem, segmental, or dispersed duplications, rather than ancient whole-genome duplication (WGD). Two gene pairs, with Ks values exceeding 2.0 (230.41 MYA and 186.75 MYA), could potentially be remnants of much older duplication events, but they are not sufficient to define a genome-wide duplication episode.

Synteny analysis of TdbZIP

To further explore the evolutionary origins of the *bZIP* gene family, we conducted collinearity analysis between *Triticum dicoccum Arabidopsis thaliana, Triticum aestivum, Hordium vulgare, Triticum turgidum subsp. Durum, and Triticum turgidum subsp. dicoccoides*. A total of 11, 11, 10,10, and 6 orthologous gene pairs of *TdbZIP* genes were identified between *T. dicoccum* and *T. aestivum, T. dicoccum* and *T. dicoccoides, T. dicoccum* and *T. durum, T. dicoccum* and *H. vulgare, T. dicoccum* and *A. thaliana*, respectively (Fig. 11). Among them, we found that *T. aestivum* and *T. dicoccoides* have the most duplicated orthologous gene pairs and few between *T. dicoccum* and *A. thaliana*, indicating closer phylogenetic relationships of *T. dicoccum* with other species. Interestingly, ten collinear gene pairs were commonly identified across the comparisons between *T. dicoccum* and *T.*

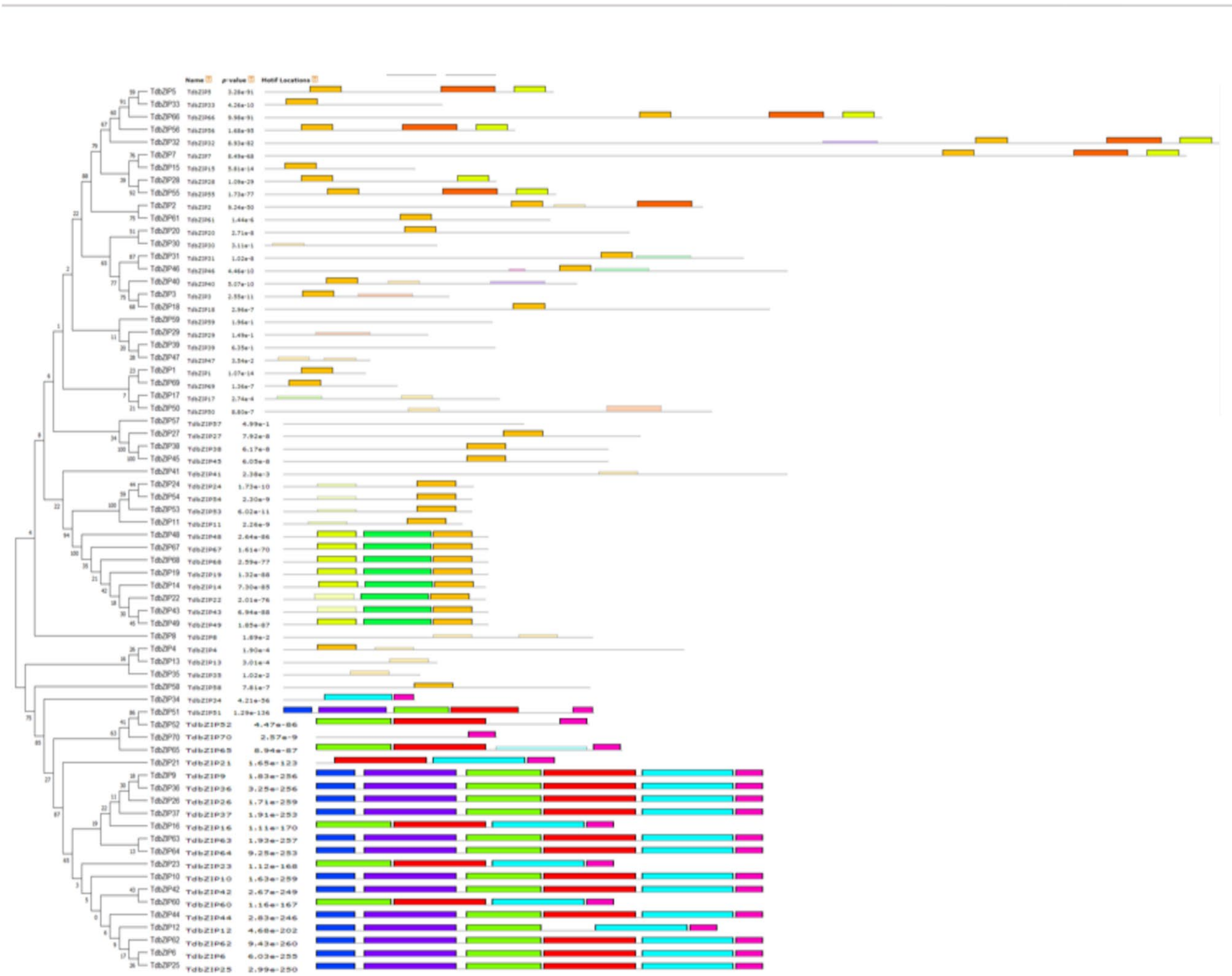

Fig. 8 Protein motifs of the *bZIP* gene family members in *T. dicoccum*. This figure illustrates the conserved motif composition across Td*bZIP* proteins, highlighting shared and unique motifs among family members

aestivum, *T. dicoccum* and *T. dicoccoides*, and *T. dicoccum* and *T. durum*. Moreover, it is noteworthy that six collinear gene pairs were shared among all five species. This widespread chromosomal distribution of collinear genes underscores the conserved evolutionary patterns and potential functional significance of the *bZIP* gene family. (Tables 2, 3, 4, 5 and 6) .

Expression patterns of *TdbZIP* genes across different wheat tissues

To investigate the tissue-specific expression of TdbZIP, expression pattern of 70 TdbZIP's was analyzed in root, flagleaf, stem, and seedling at 15 days. Genes, such as *TdbZIP*29, *TdbZIP*58, and TdbZIP39, displayed elevated expression levels in all tissue types. Root tissue generally showed high expression of bZIPs as compared to other tissues. In contrast, *TdbZIP17, TdbZIP28,* and *TdbZIP8* showed low expression in across all tissue types, suggesting that their expression might be restricted to specific developmental stages. The heatmap of the top 20 most variable genes highlighted tissue-specific expression patterns, with clear clustering between the root and aerial tissues (stem, flag leaf, and seedling) suggesting differential regulatory roles of bZIP genes in organ-specific functions, particularly root development and shoot signaling (Table 8 and Fig. 12). The clustering observed in the heatmap further emphasizes the unique expression profiles of root-specific genes, while revealing that many *TdbZIP* genes share overlapping functional roles in the flagleaf, stem, and seedling tissues. This pattern underscores the significance of *TdbZIP* genes in various developmental processes in wheat and sets the stage for further investigations into their specific biological functions and mechanisms of action.

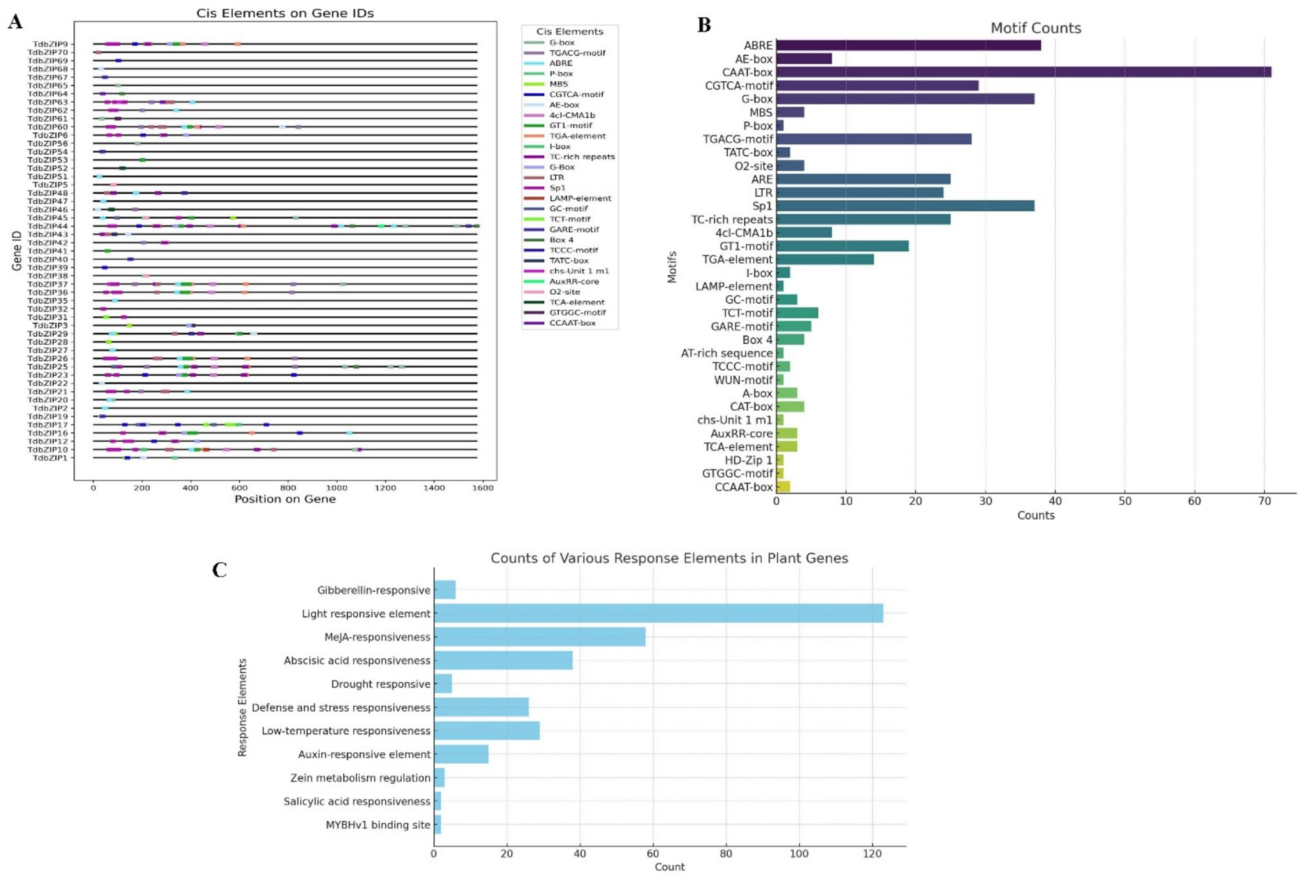

Fig. 9 **A** Cis-acting regulatory elements in TdbZIP gene promoter regions**.** **B** Frequency of cis-acting regulatory elements in TdbZIP genes **C** Frequency distribution of response elements in TdbZIP genes

Discussion

The *bZIP* TFs play important role in controlling plant growth development as well as resistance to biotic and abiotic stresses like heat, drought, salt, cold, and ABA treatment (Jakoby et al. 2002; Lee et al. 2006; Schlögl et al. 2008; Song et al. 2020; Wang et al. 2022). *bZIP* TFs have been extensively studied, such as in *Arabidopsis* (Gibalová et al. 2017), cereals like bread wheat (Song et al. 2020), rice (Wang et al. 2013), maize (Wei et al 2012a, b) *Arabidopsis* (Gibalová et al. 2017), and also in fruit crops like apple (An et al. 2018), pomegranate (Wang et al 2022), jujube (Zhang et al 2020), etc.

In this study, we identified 70 bZIP genes distributed across 14 chromosomes. The number of bZIP genes in *Triticum dicoccum* is lower than in *Arabidopsis* (78) (Dröge-Laser et al. 2018), rice (89) (Nijhawan et al. 2008), maize (125) (Wei et al. 2012a, b), and cotton (197) (Zhang et al. 2022), but higher than in tomato (69) (Li et al. 2015) and grape (55) (Liu et al. 2014). Interestingly, despite the large genome size of *Triticum dicoccum* (10.45 Gb), the number of *bZIP* genes does not scale proportionally, suggesting that gene family expansion is not solely determined by genome size. Variations in gene family size are often shaped by events such as gene duplication, deletion, pseudogenization, and functional diversification. The lower abundance of bZIP transcription factors (*TdbZIP* = 70) in *T. dicoccum* compared to hexaploid bread wheat likely contributes to its reduced adaptive potential. Higher *bZIP* copy numbers in polyploids like bread wheat enhance functional redundancy, stress buffering, and evolutionary flexibility, explaining their broader adaptation and cultivation success.

Our analysis revealed substantial sequence and structural diversity among TdbZIP genes, reflected in mRNA lengths (92–869 amino acids) and a broad range of isoelectric points (pI), implying functional and biochemical variation. Clustering in molecular weight (MW) and pI plots indicates shared biochemical properties, potentially correlating with biological roles. Genes within the MW 200–500 aa/pI 5.0–7.5 cluster were enriched for drought-responsive and ABA-responsive elements, suggesting roles in stress adaptation. Conversely, genes with higher MW (> 500 aa) and pI (> 7.0) were associated with light-responsive and gibberellin-responsive elements, implicating them in growth

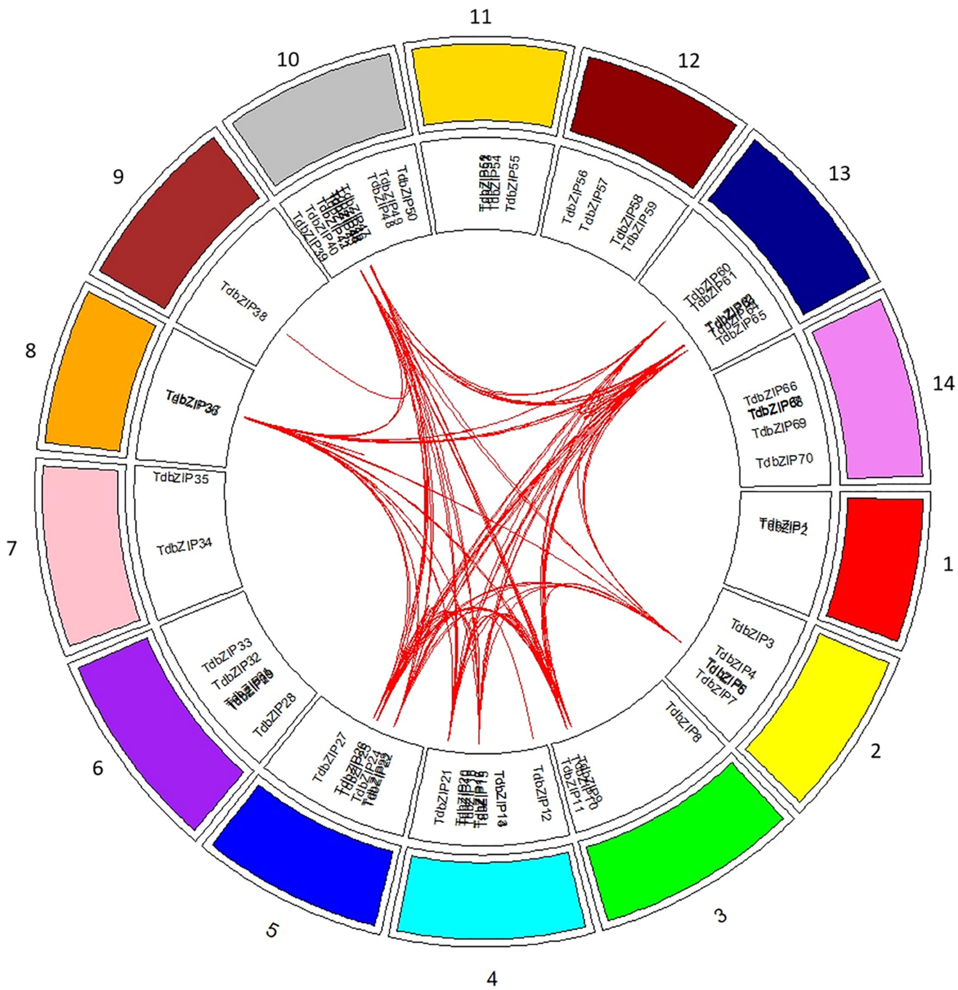

Fig. 10 The distribution of duplicated (tandem and segmental) *bZIP* genes across 14 chromosomes

and development. These functional correlations based on biophysical properties parallel observations in *Arabidopsis*, rice, and maize bZIP systems (Wei et al. 2012a, b).

Based on phylogenetic relationships and conserved motif analysis, the bZIP genes from *Triticum dicoccum* and *Arabidopsis thaliana* were classified into six subfamilies (A–F) for further analysis (Fig. 4). Subfamily B was the largest, comprising 37 TdbZIP and 10 AtbZIP genes, indicating gene expansion in *T. dicoccum*. In contrast, subfamily C included only AtbZIP genes, suggesting that this group may have been lost or functionally diverged in wheat. Subfamilies A and F showed a balanced distribution, while D and E were smaller, possibly reflecting specialized functions. Similar patterns of expansion and divergence have been observed in other crops, such as *Triticum aestivum*, where specific bZIP subfamilies are linked to stress responses (Agarwal et al. 2019).

Chromosomal localization analysis indicated that TdbZIP genes are unevenly distributed across the *Triticum dicoccum* genome, with the highest gene density observed on chromosome 10. This uneven distribution may reflect gene duplication events or chromosomal rearrangements that shaped the bZIP gene family over evolutionary time, similar to findings in other crops where bZIP genes were also found to be unevenly distributed, such as in *Triticum aestivum*, where chromosome 5A contained the largest number of bZIP genes (Liang et al. 2022a, b). The concentration of bZIP genes on certain chromosomes could have functional significance in regulating gene expression under specific developmental or environmental conditions. Structural analysis revealed diverse exon–intron organizations among TdbZIP genes, with some being intron-less and others containing up to 13 introns; this variability has been associated with complex regulatory mechanisms that enhance transcriptional flexibility, as noted in studies of bZIP genes in rice and maize (Nijhawan et al. 2008; Wei et al. 2012a, b). Also, the presence or absence of introns has been linked to stress-responsive gene upregulation, suggesting intron number contributes significantly to functional diversification within multigene families (Zhu et al 2024; Jeffares et al. 2008; Wang et al. 2013). Specifically, genes with fewer introns are often associated with rapid transcriptional activation under diverse stress conditions (Jeffares et al. 2008; Zhu et al 2024). The

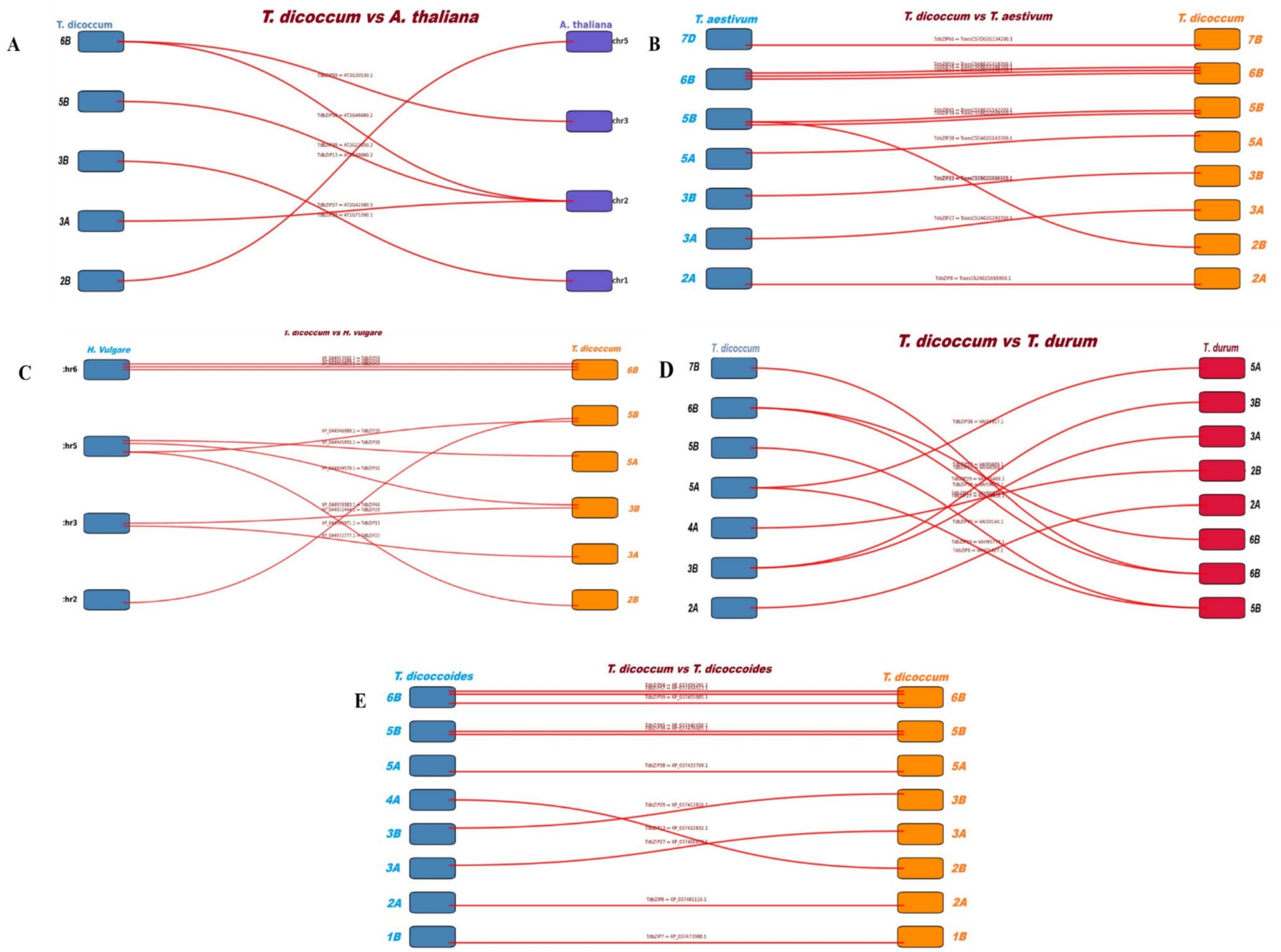

Fig. 11 Comparative synteny analysis of bZIP gene homologs between *Triticum dicoccum* and: **A** *Arabidopsis thaliana,* **B** *Triticum aestivum*, **C** *Hordeum vulgare,* **D** *Triticum turgidum subsp. dicoccoides,* **E** *Triticum turgidum subsp. durum*

Table 2 One-to-one orthologous relationships between *TdbZIP* and *Arabidopsis thaliana*

S.no	*TdbZIP* Id	Chromosome	Start	End	A. *thaliana* Id	Chromosome	Start	End
1	*TdbZIP*29	6	290,783,896	290,785,172	AT1G75390.1	1	28,291,698	28,293,008
2	*TdbZIP*39	10	8,368,021	8,368,915	AT2G22850.2	2	9,731,279	9,733,545
3	*TdbZIP*27	5	510,662,979	510,665,743	AT2G42380.5	2	17,646,885	17,649,062
4	*TdbZIP*58	12	446,248,338	446,249,570	AT2G46680.2	2	19,165,416	19,166,949
5	*TdbZIP*59	12	527,089,389	527,090,487	AT3G30530.1	3	12,139,438	12,140,293
6	*TdbZIP*13	4	296,307,744	296,308,725	AT5G48660.2	5	19,736,647	19,738,538

cis-regulatory element analysis highlighted a wide array of elements associated with light responsiveness, hormonal regulation, and stress adaptation, indicating that TdbZIP genes likely play integral roles in plant growth and stress response mechanisms, similar to the roles identified in other species (Wang et al. 2020). Predicted subcellular localization patterns indicated functional diversity among TdbZIP proteins, primarily localized in the nucleus, cytoplasm, and chloroplasts, aligning with their roles in gene regulation and metabolic pathways across different cellular compartments.

Collinearity analysis of *bZIP* transcription factor genes across emmer wheat, *Arabidopsis*, bread wheat, barley, wild emmer wheat, and durum wheat revealed a higher frequency of syntenic *TdbZIP* gene pairs between emmer wheat and its

Table 3 One-to-one orthologous relationships between *TdbZIP* and *Triticum aestivum*

S.no	*TdbZIP* Id	Chromosome	Start	End	T. *aestivum* Id	Chromosome	Start	End
1	*TdbZIP8*	3	39,066,357	39,067,450	TraesCS2A02G065900.1	4	29,812,977	29,813,664
2	*TdbZIP13*	4	296,307,744	296,308,725	TraesCS5B02G550300.1	14	701,545,510	701,549,315
3	*TdbZIP27*	5	510,662,979	510,665,743	TraesCS3A02G293700.1	7	526,508,219	526,511,244
4	*TdbZIP29*	6	290,783,896	290,785,172	TraesCS3B02G226200.1	8	313,807,932	313,808,801
5	*TdbZIP38*	9	276,878,252	276,880,943	TraesCS5A02G143300.1	13	317,693,568	317,696,195
6	*TdbZIP39*	10	8,368,021	8,368,915	TraesCS5B02G009200.1	14	9,298,645	9,299,536
7	*TdbZIP45*	10	248,381,398	248,384,055	TraesCS5B02G142200.1	14	268,112,228	268,114,952
8	*TdbZIP57*	12	230,327,705	230,328,590	TraesCS6B02G198700.1	17	237,731,023	237,731,556
9	*TdbZIP58*	12	446,248,338	446,249,570	TraesCS6B02G284300.1	17	512,447,116	512,448,413
10	*TdbZIP59*	12	527,089,389	527,090,487	TraesCS6B02G318700.1	17	566,535,116	566,536,038
11	*TdbZIP66*	14	207,879,452	207,882,732	TraesCS7D02G134200.1	21	85,213,656	85,216,297

Table 4 One-to-one orthologous relationships between *TdbZIP* and *Hordeum vulgare*

S.no	*TdbZIP Id*	Chromosome	Start	End	H. *vulgare* Id	Chromosome	Start	End
1	*TdbZIP13*	4	296,307,744	296,308,725	XP_044945971.1	5	578,583,239	578,583,524
2	*TdbZIP27*	5	510,662,979	510,665,743	XP_044972777.1	3	491,589,674	491,592,050
3	*TdbZIP29*	6	290,783,896	290,785,172	XP_044972444.1	3	294,386,163	294,386,630
4	*TdbZIP32*	6	425,122,734	425,126,532	XP_044949570.1	5	298,226,573	298,227,261
5	*TdbZIP38*	9	276,878,252	276,880,943	XP_044945955.1	5	316,109,937	316,112,203
6	*TdbZIP39*	10	8,368,021	8,368,915	XP_044946989.1	5	3,719,451	3,720,095
7	*TdbZIP44*	10	243,339,050	243,342,127	XP_044970383.1	2	571,963,342	571,963,986
8	*TdbZIP57*	12	230,327,705	230,328,590	XP_044952813.1	6	161,297,685	161,298,239
9	*TdbZIP58*	12	446,248,338	446,249,570	XP_044955584.1	6	401,468,682	401,469,442
10	*TdbZIP59*	12	527,089,389	527,090,487	XP_044953502.1	6	487,722,262	487,722,852

Table 5 One-to-one orthologous relationships between *TdbZIP* and *Triticum turgidum* subsp. *dicoccoides*

S.no	*TdbZIP* Id	Chromosome	Start	End	T. *dicoccoides* Id	Chromosome	Start	End
1	*TdbZIP7*	2	525,074,355	525,078,162	XP_037473988.1	2	44,017,777	44,017,693
2	*TdbZIP8*	3	39,066,357	39,067,450	XP_037481110.1	3	37,422,047	37,422,662
3	*TdbZIP13*	4	296,307,744	296,308,725	XP_037422832.1	7	618,279,924	618,277,512
4	*TdbZIP27*	5	510,662,979	510,665,743	XP_037406679.1	5	536,990,543	536,992,796
5	*TdbZIP29*	6	290,783,896	290,785,172	XP_037412816.1	6	320,776,249	320,776,698
6	*TdbZIP38*	9	276,878,252	276,880,943	XP_037433709.1	9	263,508,249	263,510,585
7	*TdbZIP39*	10	8,368,021	8,368,915	XP_037439005.1	10	8,840,564	8,841,199
8	*TdbZIP45*	10	248,381,398	248,384,055	XP_037440450.1	10	281,510,254	281,512,546
9	*TdbZIP57*	12	230,327,705	230,328,590	XP_037454513.1	12	247,943,712	247,944,245
10	*TdbZIP58*	12	446,248,338	446,249,570	XP_037456291.1	12	481,154,251	481,155,031
11	*TdbZIP59*	12	527,089,389	527,090,487	XP_037455885.1	12	565,533,636	565,534,259

close relatives bread wheat and wild emmer wheat than with *Arabidopsis*. This pattern suggests lineage-specific expansion of the *bZIP* gene family during emmer wheat evolution, likely following its divergence from the *Arabidopsis* lineage. Furthermore, several *TdbZIP* genes exhibited multiple syntenic relationships across species, indicating that they may represent conserved ancestral genes retained during speciation and potentially pivotal in the evolutionary trajectory of wheat *bZIP* genes (Cannon et al. 2004).

To assess selective pressures acting on the *TdbZIP* genes, we calculated the ratio of non-synonymous (Ka) to synonymous (Ks) substitution rates (Ka/Ks) for all paralogous

Table 6 One-to-one orthologous relationships between *TdbZIP* and *Triticum turgidum subsp. durum*

S.no	*TdbZIP* Id	Chromosome	Start	End	T. *durum*	Chromosome	Start	End
1	*TdbZIP8*	3	39,066,357	39,067,450	VAH25427.1	3	28,896,859	28,897,081
2	*TdbZIP13*	4	296,307,744	296,308,725	VAH96838.1	7	611,888,740	611,889,198
3	*TdbZIP27*	5	510,662,979	510,665,743	VAH79818.1	6	528,163,533	528,165,756
4	*TdbZIP29*	6	290,783,896	290,785,172	VAH76468.1	6	315,157,073	315,157,522
5	*TdbZIP38*	9	276,878,252	276,880,943	VAI15917.1	9	311,092,316	311,094,651
6	*TdbZIP39*	10	8,368,021	8,368,915	VAH95778.1	9	586,072,140	586,072,634
7	*TdbZIP45*	10	248,381,398	248,384,055	VAI30144.1	10	267,274,731	267,277,027
8	*TdbZIP57*	12	230,327,705	230,328,590	VAI90489.1	14	514,524,638	514,525,141
9	*TdbZIP58*	12	446,248,338	446,249,570	VAI59015.1	12	459,475,818	459,476,598
10	*TdbZIP59*	12	527,089,389	527,090,487	VAI60388.1	12	542,515,445	542,516,068

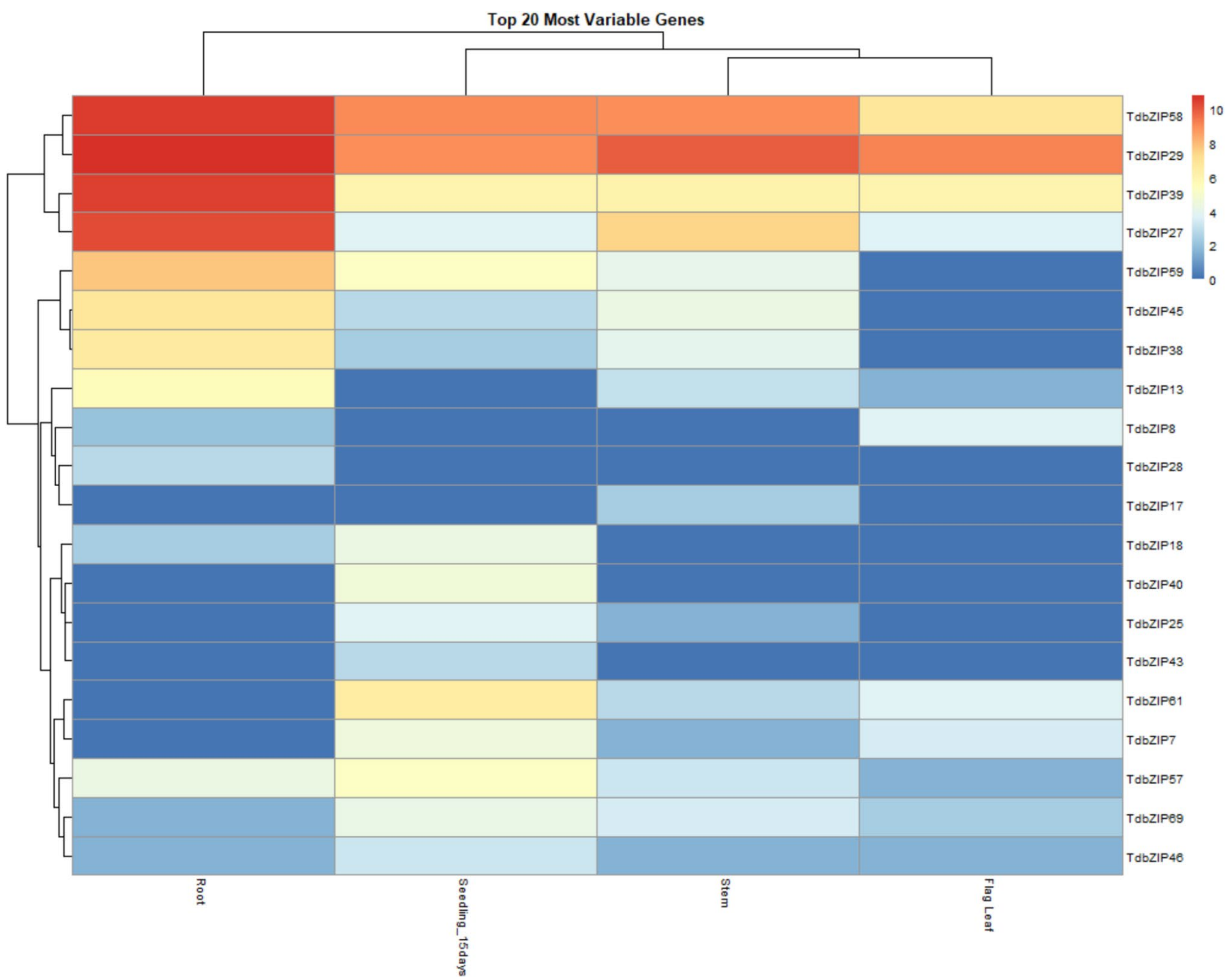

Fig. 12 Expression patterns of TdbZIP genes across various tissues in emmer wheat

gene pairs. The Ka/Ks ratios for all pairs were significantly less than 1, indicative of strong purifying selection throughout emmer wheat evolution. This suggests that these transcription factors have experienced significant evolutionary constraint, with limited functional divergence among paralogs (Cannon et al. 2004; Liu et al. 2023).

Table 7 Pairwise evolutionary analysis of bZIP transcription factors in *T. dicoccum* based on Ka, Ks, and divergence estimates

S.no	Seq_1	bZIP Id	Seq_2	bZIP Id	Ka	Ks	Divergence time (MYA)
1	CM_00002.g54584.t1	*TdbZIP*5	CM_00006.g68736.t1	*TdbZIP*33	0.070724	1.1508	88.5262
2	CM_00002.g65881.t1	*TdbZIP*7	CM_00004.g54147.t1	*TdbZIP*15	0.174578	1.1133	85.6372
3	CM_00006.g10718.t1	*TdbZIP*28	CM_00011.g55869.t1	*TdbZIP*55	0.140427	0.7191	55.3152
4	CM_00001.g20965.t1	*TdbZIP*2	CM_00013.g41461.t1	*TdbZIP*61	0.518538	2.3274	179.0321
5	CM_00004.g69497.t1	*TdbZIP*20	CM_00006.g38243.t1	*TdbZIP*30	0.610382	0.824	63.3815
6	CM_00006.g39817.t1	*TdbZIP*31	CM_00010.g34760.t1	*TdbZIP*46	0.311413	0.6128	47.1354
7	CM_00002.g17164.t1	*TdbZIP*3	CM_00004.g64415.t1	*TdbZIP*18	0.220775	0.7234	55.6485
8	CM_00010.g795.t1	*TdbZIP*39	CM_00010.g39821.t1	*TdbZIP*47	0.883179	2.9953	230.4078
9	CM_00001.g18149.t1	*TdbZIP*1	CM_00014.g52546.t1	*TdbZIP*69	0.84304	2.4277	186.7498
10	CM_00004.g56812.t1	*TdbZIP*17	CM_00010.g80651.t1	*TdbZIP*50	0.931114	–	–
11	CM_00009.g33392.t1	*TdbZIP*38	CM_00010.g31061.t1	*TdbZIP*45	0.003658	0.0489	3.7632
12	CM_00005.g32783.t1	*TdbZIP*24	CM_00011.g43005.t1	*TdbZIP*54	0.072677	0.0861	6.6259
13	CM_00010.g26988.t1	*TdbZIP*43	CM_00010.g67986.t1	*TdbZIP*49	0.09618	0.1831	14.0821
14	CM_00002.g42574.t1	*TdbZIP*4	CM_00004.g37463.t1	*TdbZIP*13	0.871663	–	–
15	CM_00011.g35995.t1	*TdbZIP*51	CM_00011.g35999.t1	*TdbZIP*52	0.046459	0.0585	4.5007
16	CM_00003.g83892.t1	*TdbZIP*9	CM_00008.g42268.t1	*TdbZIP*36	0.008764	0.0688	5.2905
17	CM_00013.g61048.t1	*TdbZIP*63	CM_00013.g66005.t1	*TdbZIP*64	0.012274	0.1457	11.2104
18	CM_00010.g20308.t1	*TdbZIP*42	CM_00013.g34169.t1	*TdbZIP*60	0.026221	0.0894	6.8775
19	CM_00002.g55552.t1	*TdbZIP*6	CM_00005.g41163.t1	*TdbZIP*25	0.015832	0.0986	7.586

The tissue-specific expression analysis of *TdbZIP* genes in *Triticum dicoccum* revealed distinct patterns across root, flag leaf, stem, and seedling tissues. Several genes, like *TdbZIP29, TdbZIP58, TdbZIP39,* and *TdbZIP27*, showed elevated expression in all tissues, suggesting that they might have fundamental role in plant development, metabolism and various stresses. Root tissues displayed generally higher expression levels suggesting their crucial roles in processes such as nutrient absorption, root architecture, overall root development and likely regulate stress-responsive pathways.

Conclusion

Our genome-wide identification and in-depth analysis of the *bZIP* gene family in *Triticum dicoccum* offers a foundational understanding of this family's structural, evolutionary, and functional characteristics. We identified 70 Td*bZIP* genes with varied structural features, chromosomal distributions, and distinct expression profiles, which point to their functional diversity and potential regulatory roles in development and stress responses.

The absence of certain *bZIP* groups, along with the uneven chromosomal distribution and distinct expression patterns across tissues, suggests an adaptation of the *bZIP* family to meet the physiological demands of *T. dicoccum* in diverse environmental contexts. The findings from cis-regulatory element analysis support the hypothesis that Td*bZIP* genes play key roles in mediating responses to light, hormones, and stress, underscoring their potential significance in plant adaptation to abiotic stresses.

In conclusion, our work not only enhances the current understanding of the *bZIP* gene family in wheat but also lays the groundwork for future research aimed at unraveling the specific biological functions and mechanisms of Td*bZIP* genes. Given the critical roles of *bZIP* genes in growth and stress responses, functional studies on candidate genes identified in this study, particularly those with high tissue-specific expression, could reveal valuable insights for breeding strategies focused on improving stress resilience and productivity in wheat. Future research involving functional characterization of Td*bZIP* genes, gene-editing approaches, and gene expression analysis under diverse environmental conditions will further elucidate the regulatory networks governing wheat's adaptive and developmental processes.

Table 8 Top 20 differentially expressed *TdbZIP* genes (|log2FC|> 2) in *T. dicoccum*

Contrast	GeneID	log2 fold change	*P* value
Flag Leaf_vs_Root_DEGs	*TdbZIP*13	−4.030	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*18	−2.322	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*2	−2.322	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*27	−6.596	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*28	−2.807	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*38	−6.570	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*39	−4.366	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*45	−6.714	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*57	−2.737	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*58	−3.843	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*59	−7.775	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*61	3.700	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*67	−2.322	0.01
Flag Leaf_vs_Root_DEGs	*TdbZIP*7	3.459	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*18	−4.248	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*25	−3.700	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*31	−2.322	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*38	−2.322	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*4	−2.322	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*40	−4.644	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*43	−2.807	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*45	−2.807	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*57	−3.624	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*58	−2.286	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*59	−5.285	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*61	−2.775	0.01
Flag Leaf_vs_Seedling_15days_DEGs	*TdbZIP*8	3.700	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*13	−5.615	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*2	−2.322	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*25	3.700	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*27	−6.481	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*28	−2.807	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*31	2.322	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*38	−4.248	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*39	−4.366	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*40	4.644	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*43	2.807	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*45	−3.907	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*59	−2.489	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*61	6.476	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*69	2.585	0.01
Seedling_15days_vs_Root_DEGs	*TdbZIP*7	4.524	0.01
Stem_vs_Flag Leaf_DEGs	*TdbZIP*17	2.322	0.01
Stem_vs_Flag Leaf_DEGs	*TdbZIP*27	3.824	0.01
Stem_vs_Flag Leaf_DEGs	*TdbZIP*38	3.907	0.01
Stem_vs_Flag Leaf_DEGs	*TdbZIP*45	4.322	0.01
Stem_vs_Flag Leaf_DEGs	*TdbZIP*58	2.144	0.01
Stem_vs_Flag Leaf_DEGs	*TdbZIP*59	4.087	0.01
Stem_vs_Flag Leaf_DEGs	*TdbZIP*8	−3.700	0.01
Stem_vs_Root_DEGs	*TdbZIP*13	−2.615	0.01
Stem_vs_Root_DEGs	*TdbZIP*17	2.322	0.01

Table 8 (continued)

Contrast	GeneID	log2 fold change	*P* value
Stem_vs_Root_DEGs	*TdbZIP*18	−2.322	0.01
Stem_vs_Root_DEGs	*TdbZIP*2	−2.322	0.01
Stem_vs_Root_DEGs	*TdbZIP*27	−2.772	0.01
Stem_vs_Root_DEGs	*TdbZIP*28	−2.807	0.01
Stem_vs_Root_DEGs	*TdbZIP*38	−2.663	0.01
Stem_vs_Root_DEGs	*TdbZIP*39	−4.301	0.01
Stem_vs_Root_DEGs	*TdbZIP*45	−2.392	0.01
Stem_vs_Root_DEGs	*TdbZIP*59	−3.687	0.01
Stem_vs_Root_DEGs	*TdbZIP*61	2.807	0.01
Stem_vs_Root_DEGs	*TdbZIP*67	−2.322	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*13	3.000	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*17	2.322	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*18	−4.248	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*25	−2.115	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*27	3.709	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*31	−2.322	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*4	−2.322	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*40	−4.644	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*43	−2.807	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*57	−2.040	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*61	−3.668	0.01
Stem_vs_Seedling_15days_DEGs	*TdbZIP*7	−2.939	0.01

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Author contributions DCM and AKS conceived and designed the study. LDP performed the data curation, analysis, and wrote the first draft of the manuscript. DCM and AKS carried out formal analysis, reviewed and edited the manuscript. SDM and ST contributed to methodology, supervision, reviewed and edited the manuscript. KKC, SBL, AS, SK, and SSJ provided overall guidance and finalized the edited manuscript. NB, MK, and GKJ contributed formal analysis and reviewed the manuscript. All authors contributed to the article and approved the final manuscript.

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Data availability The raw data PacBio HiFi reads have been generated at the National Bureau of Plant Genetic Resources (NBPGR), New Delhi, India (110,012). These datasets can be accessed upon request through the NBPGR.

Declarations

Conflict of interest The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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References

Aboyoun P, Pages H, Lawrence M (2013) GenomicRanges: representation and manipulation of genomic intervals. R Package Version 1(12):5

Agarwal P, Baranwal VK, Khurana P (2019) Genome-wide analysis of bZIP transcription factors in wheat and functional characterization of a TabZIP under abiotic stress. Sci Rep 9(1):4608

An JP, Yao JF, Xu RR, You CX, Wang XF, Hao YJ (2018) Apple bZIP transcription factor MdbZIP44 regulates abscisic acid-promoted anthocyanin accumulation. Plant Cell Environ 41(11):2678–2692

Andrews S (2010) FastQC: a quality control tool for high throughput sequence data. Available online at: http://www.bioinformatics.babraham.ac.uk/projects/fastqc

Bailey TL, Boden M, Buske FA, Frith M, Grant CE, Clementi L, Noble WS (2015) MEME suite: tools for motif discovery and searching. Nucleic Acids Res 43(W1):W39–W49

Bi C, Yu Y, Dong C, Yang Y, Zhai Y, Du F et al (2021) The *bZIP* transcription factor Ta*bZIP*15 improves salt stress tolerance in wheat. Plant Biotechnol J 19(2):209–211

Bolger AM, Lohse M, Usadel B (2014) Trimmomatic: a flexible trimmer for Illumina sequence data. Bioinformatics 30(15):2114–2120

Buchfink B, Xie C, Huson DH (2015) Fast and sensitive protein alignment using DIAMOND. Nat Methods 12(1):59–60

Cannon SB, Mitra A, Baumgarten A, Young ND, May G (2004) The roles of segmental and tandem gene duplication in the evolution of large gene families in *Arabidopsis thaliana*. BMC Plant Biol 4:1–21

Chao J, Li Z, Sun Y, Aluko OO, Wu X, Wang Q, Liu G (2021) MG2C: a user-friendly online tool for drawing genetic maps. Mol Hortic 1:1–4

Chen C, Chen H, Zhang Y, Thomas HR, Frank MH, He Y, Xia R (2020) TBtools: an integrative toolkit developed for interactive analyses of big biological data. Mol Plant 13(8):1194–1202

Dong W, Xie Q, Liu Z, Han Y, Wang X, Xu R, Gao C (2023) Genome-wide identification and expression profiling of the bZIP gene family in *Betula platyphylla* and the functional characterization of BpChr04G00610 under low-temperature stress. Plant Physiol Biochem 198:107676

Dröge-Laser W, Snoek BL, Snel B, Weiste C (2018) The Arabidopsis bZIP transcription factor family—an update. Curr Opin Plant Biol 45:36–49

Duan L, Mo Z, Fan Y, Li K, Yang M, Li D et al (2022) Genome-wide identification and expression analysis of the *bZIP* transcription factor family genes in response to abiotic stress in *Nicotiana tabacum* L. BMC Genomics 23(1):318

Eddy SR (2011) Accelerated profile HMM searches. PLoS Comput Biol 7(10):e1002195

Edgar RC (2004) MUSCLE: multiple sequence alignment with high accuracy and high throughput. Nucleic Acids Res 32(5):1792–1797

Fadida-Myers A, Fuerst D, Tzuberi A, Yadav S, Nashef K, Roychowdhury R, Ben-David R (2022) Emmer wheat eco-geographic and genomic congruence shapes phenotypic performance under Mediterranean climate. Plants 11(11):1460

Gai W, Ma X, Qiao Y, Shi B, Ul Haq S, Li Q et al (2020) Characterization of the bZIP transcription factor family in pepper (Capsicum annuum L.): CabZIP25 positively modulates salt tolerance. Front Plant Sci 11:139

Gibalová A, Steinbachová L, Hafidh S, Bláhová V, Gadiou Z, Michailidis C et al (2017) Characterization of pollen-expressed *bZIP* protein interactions and the role of AT*bZIP*18 in the male gametophyte. Plant Reprod 30(1):1–17

Gu Z, Gu L, Eils R, Schlesner M, & Brors B (2014) "Circlize" implements and enhances circular visualization in R.

Guan Y, Ren H, Xie H, Ma Z, Chen F (2009) Identification and characterization of *bZIP*-type transcription factors involved in carrot (*Daucus carota* L.) somatic embryogenesis. Plant J 60(2):207–217

Horton P, Park KJ, Obayashi T, Fujita N, Harada H, Adams-Collier CJ, Nakai K (2007) WoLF PSORT: protein localization predictor. Nucleic Acids Res 35(Suppl 2):W585–W587

Hu B, Jin J, Guo AY, Zhang H, Luo J, Gao G (2015) GSDS 2.0: an upgraded gene feature visualization server. Bioinformatics 31(8):1296–1297

Jakoby M, Weisshaar B, Dröge-Laser W, Vicente-Carbajosa J, Tiedemann J, Kroj T, et al (2002) *bZIP* transcription factors in Arabidopsis. Trends Plant Sci 7(3):106–111

Jeffares DC, Penkett CJ, Bähler J (2008) Rapidly regulated genes are intron poor. Trends Genet 24(8):375–378

Lawrence M, Gentleman R, Carey V (2009) Rtracklayer: an R package for interfacing with genome browsers. Bioinformatics 25(14):1841

Lee SC, Choi HW, Hwang IS, Choi DS, Hwang BK (2006) Functional roles of the pepper pathogen-induced *bZIP* transcription factor, CA*bZIP*1, in enhanced resistance to pathogen infection and environmental stresses. Planta 224:1209–1225

Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouzé P, Rombauts S (2002) PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in silico analysis of promoter sequences. Nucleic Acids Res 30(1):325–327

Letunic I, Bork P (2021) Interactive tree of life (iTOL) v5: an online tool for phylogenetic tree display and annotation. Nucleic Acids Res 49(W1):W293–W296

Letunic I, Khedkar S, Bork P (2021) SMART: Recent updates, new developments and status in 2020. Nucleic Acids Res 49(D1):D458–D460

Li H, Durbin R (2009) Fast and accurate short read alignment with burrows-wheeler transform. Bioinformatics 25(14):1754–1760

Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, Homer N, Marth G, Abecasis G, Durbin R (2009) The sequence alignment/map format and SAMtools. Bioinformatics 25(16):2078–2079

Li D, Fu F, Zhang H, Song F (2015) Genome-wide systematic characterization of the bZIP transcriptional factor family in tomato (*Solanum lycopersicum* L.). BMC Genomics 16:1–18

Li Y, Meng D, Li M, Cheng L (2016) Genome-wide identification and expression analysis of the *bZIP* gene family in apple (*Malus domestica*). Tree Genet Genomes 12:1–17

Liang Y, Xia J, Jiang Y, Bao Y, Chen H, Wang D et al (2022a) Genome-wide identification and analysis of *bZIP* gene family and resistance of TaABI5 (Ta*bZIP*96) under freezing stress in wheat (*Triticum aestivum*). Int J Mol Sci 23(4):235

Liang Y, Xia J, Jiang Y, Bao Y, Chen H, Wang D, Zhang Da, Yu J, Cang J (2022b) Genome-wide identification and analysis of bZIP gene family and resistance of TaABI5 (TabZIP96) under freezing stress in wheat (*Triticum aestivum*). Int J Mol Sci 23(4):2351

Liu JX, Srivastava R, Che P, Howell SH (2007) An endoplasmic reticulum stress response in Arabidopsis is mediated by proteolytic processing and nuclear relocation of a membrane-associated transcription factor, bZIP28. Plant Cell 19(12):4111–4119

Liu J, Chen N, Chen F, Cai B, Dal Santo S, Tornielli GB, Cheng ZM (2014) Genome-wide analysis and expression profile of the bZIP transcription factor gene family in grapevine (Vitis vinifera). BMC Genomics 15:1–18

Liu W, Wang M, Zhong M, Luo C, Shi S, Qian Y, Kang Y, Jiang B (2023) Genome-wide identification of bZIP gene family and expression analysis of BhbZIP58 under heat stress in wax gourd. BMC Plant Biol 23(1):598

Ma H, Liu C, Li Z, Ran Q, Xie G, Wang B et al (2018) Zm*bZIP*4 contributes to stress resistance in maize by regulating ABA synthesis and root development. Plant Physiol 178(2):753–770

Ma M, Chen Q, Dong H, Zhang S, Huang X (2021) Genome-wide identification and expression analysis of the *bZIP* transcription factors, and functional analysis in response to drought and cold stresses in pear (*Pyrus bretschneideri*). BMC Plant Biol 21(1):583

Madival SD, Mishra DC, Chaturvedi KK, Sharma A, Budhlakoti N, Angadi UB, Basavaraja P, Farooqi MS, Srivastava S, Jha GK (2024) NaturePred: a tool for revolutionizing natural product classification with artificial intelligence. Curr Proteomics 21(5):429–436

Nijhawan A, Jain M, Tyagi AK, Khurana JP (2008) Genomic survey and gene expression analysis of the basic leucine zipper transcription factor family in rice. Plant Physiol 146(2):333

Punta M, Coggill PC, Eberhardt RY, Mistry J, Tate J, Boursnell C, Pang N, Forslund K, Ceric G, Clements J, Heger A, Holm

L, Sonnhammer ELL, Eddy SR, Bateman A, Finn RD (2012) The Pfam protein families database. Nucleic Acids Res 40(D1):D290–D301

Schlögl PS, Nogueira FTS, Drummond R, Felix JM, De Rosa VE, Vicentini R, Menossi M (2008) Identification of new ABA-and MEJA-activated sugarcane bZIP genes by data mining in the SUCEST database. Plant Cell Rep 27:335–345

Song Y, Luo G, Shen L, Yu K, Yang W, Li X, Sun J, Zhan K, Cui D, Liu D, Zhang A (2020) TubZIP28, a novel bZIP family transcription factor from *Triticum urartu*, and TabZIP28, its homologue from *Triticum aestivum*, enhance starch synthesis in wheat. New Phytol 226(5):1384–1398

Strathmann A, Kuhlmann M, Heinekamp T, Dröge-Laser W (2001) BZI-1 specifically heterodimerises with the tobacco *bZIP* transcription factors BZI-2, BZI-3/TBZF, and BZI-4, and is functionally involved in flower development. Plant J 28(4):397–408

Tamura K, Stecher G, Kumar S (2021) MEGA11: molecular evolutionary genetics analysis version 11. Mol Biol Evol 38(7):3022–3027

Toh S, McCourt P, Tsuchiya Y (2012) HY5 is involved in strigolactone-dependent seed germination in Arabidopsis. Plant Signal Behav 7(5):556–558

Wang JC, Xu H, Zhu Y, Liu QQ, Cai XL (2013) OsbZIP58, a basic leucine zipper transcription factor, regulates starch biosynthesis in rice endosperm. J Exp Bot 64(11):3453–3466

Wang X, Lu X, Malik WA, Chen X, Wang J, Wang D, Wang S, Chen C, Guo L, Ye W (2020) Differentially expressed bZIP transcription factors confer multi-tolerances in *Gossypium hirsutum* L. Int J Biol Macromol 146:569–578

Wang S, Zhang X, Li B, Zhao X, Shen Y, Yuan Z (2022) Genome-wide identification and characterization of the *bZIP* gene family and cloning of candidate genes for anthocyanin biosynthesis in pomegranate (*Punica granatum*). BMC Plant Biol 22(1):1–18

Wei K, Chen J, Wang Y, Chen Y, Chen S, Lin Y et al (2012a) Genome-wide analysis of *bZIP*-encoding genes in maize. DNA Res 19(6):463–476

Wei KAIFA, Chen JUAN, Wang Y, Chen Y, Chen S, Lin Y, Wei K, Chen J, Pan S, Zhong X, Xie D (2012b) Genome-wide analysis of bZIP-encoding genes in maize. DNA Res 19(6):463–476

Weiste C, Pedrotti L, Selvanayagam J, Muralidhara P, Fröschel C, Novák O et al (2017) The Arabidopsis *bZIP*11 transcription factor links low-energy signalling to auxin-mediated control of primary root growth. PLoS Genet 13(2):e1006607

Wu S, Zhu P, Jia B, Yang J, Shen Y, Cai X et al (2018) A *Glycine soja* group S2 *bZIP* transcription factor Gs*bZIP*67 conferred bicarbonate alkaline tolerance in *Medicago sativa*. BMC Plant Biol 18(1):234

Yoshida T, Fujita Y, Sayama H, Kidokoro S, Maruyama K, Mizoi J et al (2010) AREB1, AREB2, and ABF3 are master transcription factors that cooperatively regulate ABRE-dependent ABA signaling involved in drought stress tolerance and require ABA for full activation. Plant J 61(4):672–685

Zhang Y, Gao W, Li H, Wang Y, Li D, Xue C, Zhao J (2020) Genome-wide analysis of the bZIP gene family in Chinese jujube (Ziziphus jujuba Mill.). BMC Genomics 21:1–14

Zhang B, Feng C, Chen L, Li B, Zhang X, Yang X (2022) Identification and functional analysis of bZIP genes in cotton response to drought stress. Int J Mol Sci 23(23):14894

Zhu X, Gao T, Bian K, Meng C, Tang X, Mao Y (2024) Genome-wide analysis and expression profile of the bZIP gene family in *Neopyropia yezoensis*. Front Plant Sci 15:1461922